INCH-POUND

MIL-M-28787/358
26 October 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

DMA SUPPORT MODULE, KEY CODE FUN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic Direct Memory Access (DMA) support module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.3034 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be FUN.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 120 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE 1. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
A2-A13	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B5-B8	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	---------Module shall pass all remaining tests---------						
D1-D38	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D39-D41	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D42	Ibd	V	S	0.0	0.3	0.0	0.3	0.0	0.3	mA
D43-D58	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D59-D61	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D62	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	mA
E1-E62	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
F1-F4	Ios	VII	S	-100.0	-15.0	-100.0	-15.0	-100.0	-15.0	mA
F5-F36	Ios	VII	S	-225.86	-40.87	-225.86	-40.87	-225.86	-40.87	mA
F37-F42	Ios	VII	S	-130.0	-30.0	-130.0	-30.0	-130.0	-30.0	mA
F43-F48	Ios	VII	S	-250.0	-30.0	-250.0	-30.0	-250.0	-30.0	mA
F49,F50	Ios	VII	S	-130.27	-30.0	-130.27	-30.0	-130.27	-30.0	mA
F51	Ios	VII	S	-250.58	-30.52	-250.58	-30.52	-250.58	-30.52	mA
G1,G2	Iih	VIII	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
G3-G34	Iih	VIII	X	-508.0	-421.5	-508.0	-421.5	-508.0	-421.5	uA
G35-G38	Iih	VIII	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
G39-G41	Iih	VIII	X	0.0	40.0	0.0	40.0	0.0	40.0	uA
G42	Iih	VIII	X	0.0	60.0	0.0	60.0	0.0	60.0	uA
G43-G58	Iih	VIII	X	0.0	25.0	0.0	25.0	0.0	25.0	uA
G59-G61	Iih	VIII	X	0.0	50.0	0.0	50.0	0.0	50.0	uA
G62	Iih	VIII	X	0.0	75.0	0.0	75.0	0.0	75.0	uA

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
H1,H2	Iil	IX	X	-0.4	-0.03	-0.4	-0.03	-0.4	-0.03	mA
H3-H34	Iil	IX	X	-1.09	-0.804	-1.09	-0.804	-1.09	-0.804	mA
H35-H38	Iil	IX	X	-200.0	0.0	-200.0	0.0	-200.0	0.0	uA
H39-H41	Iil	IX	X	-0.8	-0.06	-0.8	-0.06	-0.8	-0.06	mA
H42	Iil	IX	X	-1.2	-0.06	-1.2	-0.06	-1.2	-0.06	mA
H43-H58	Iil	IX	X	-0.25	0.0	-0.25	0.0	-0.25	0.0	mA
H59-H61	Iil	IX	X	-0.5	0.0	-0.5	0.0	-0.5	0.0	mA
H62	Iil	IX	X	-0.75	0.0	-0.75	0.0	-0.75	0.0	mA
J1-J4	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J5-J51	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J52	Voh	X	X	2.4	2.6	2.4	2.6	2.4	2.6	V
K1-K36	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K37-K51	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
L1-L32	Iozh	XII	X	-500.0	-421.5	-500.0	-421.5	-500.0	-421.5	uA
L33-L35	Iozh	XII	X	-100.0	100.0	-100.0	100.0	-100.0	100.0	uA
L36	Iozh	XII	X	-426.3	-195.2	-426.3	-195.2	-426.3	-195.2	uA
M1-M32	Iozl	XIII	X	-1.09	-0.804	-1.09	-0.804	-1.09	-0.804	uA
M33-M35	Iozl	XIII	X	-100.0	100.0	-100.0	100.0	-100.0	100.0	uA
M36	Iozl	XIII	X	-637.0	-396.0	-637.0	-396.0	-637.0	-396.0	uA
N	Icc	XIV	X	59	596	59	596	59	596	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	O	See table XVI						
R	PSFC	XVIII	S	3.7	6.9	3.7	6.9	3.7	6.9	uF
S	PSTA	XIX	O	0	400	0	400	0	400	mV
T1-T15	Output crosstalk (high level)	XX	O	0	500	0	500	0	500	mV

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
T16-T19	Output crosstalk (high level)	XX	Q	0	400	0	400	0	400	mV
T20-T51	Output crosstalk (high level)	XX	Q	0	525	0	525	0	525	mV
U1-U12	Output crosstalk (low level)	XXI	Q	0	500	0	500	0	500	mV
U13	Output crosstalk (low level)	XXI	Q	0	800	0	800	0	800	mV
U14	Output crosstalk (low level)	XXI	Q	0	500	0	500	0	500	mV
U15	Output crosstalk (low level)	XXI	Q	0	800	0	800	0	800	mV
U16-U19	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV
U20-U51	Output crosstalk (low level)	XXI	Q	0	925	0	925	0	925	mV
V1-V32	Crosstalk duration	XXII	Q	0	61	0	61	0	61	nS
V33,V34	Crosstalk duration	XXII	Q	0	45	0	45	0	45	nS
W1-W38	Cin	XXIII	Q	2	45	2	45	2	45	pF
W39-W43	Cin	XXIII	Q	2	35	2	35	2	85	pF
W44-W55	Cin	XXIII	Q	2	25	2	25	2	25	pF
W56-W60	Cin	XXIII	Q	2	55	2	55	2	55	pF
W61,W62	Cin	XXIII	Q	2	85	2	85	2	85	pF

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE 1. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
X1-X64	Disable time test	XXIV	Q	2.0	39.0	2.0	39.0	2.0	39.0	nS
X65	Disable time test	XXIV	Q	0.0	33.0	0.0	33.0	0.0	33.0	nS
X66-X68	Disable time test	XXIV	Q	0.0	32.0	0.0	32.0	0.0	32.0	nS
X69	Disable time test	XXIV	Q	0.0	43.0	0.0	43.0	0.0	43.0	nS
X70-X72	Disable time test	XXIV	Q	0.0	46.0	0.0	46.0	0.0	46.0	nS
Y	Threshold	XXV	X	See table XXV						

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/358-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

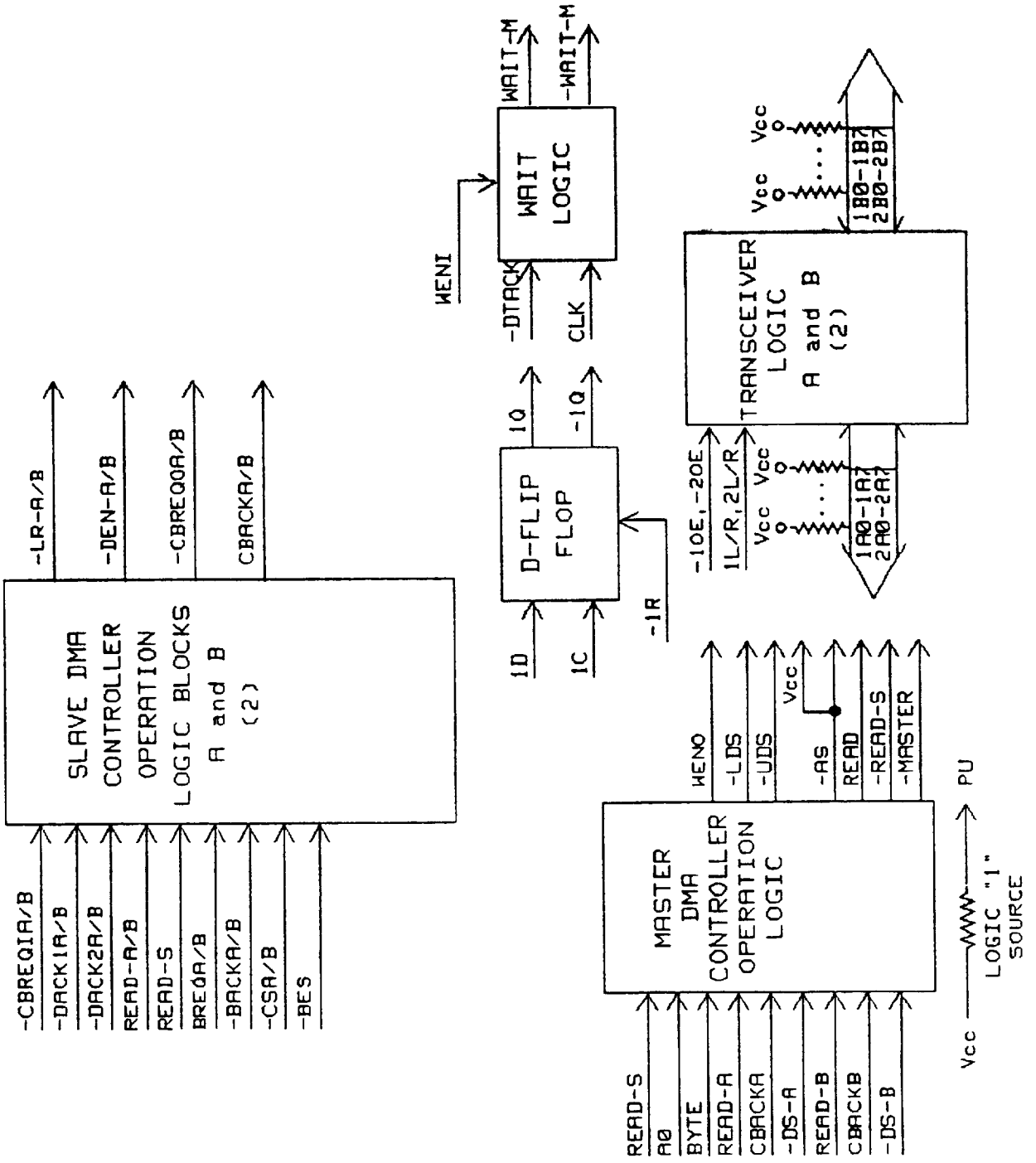


FIGURE 1. FUN functional block diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

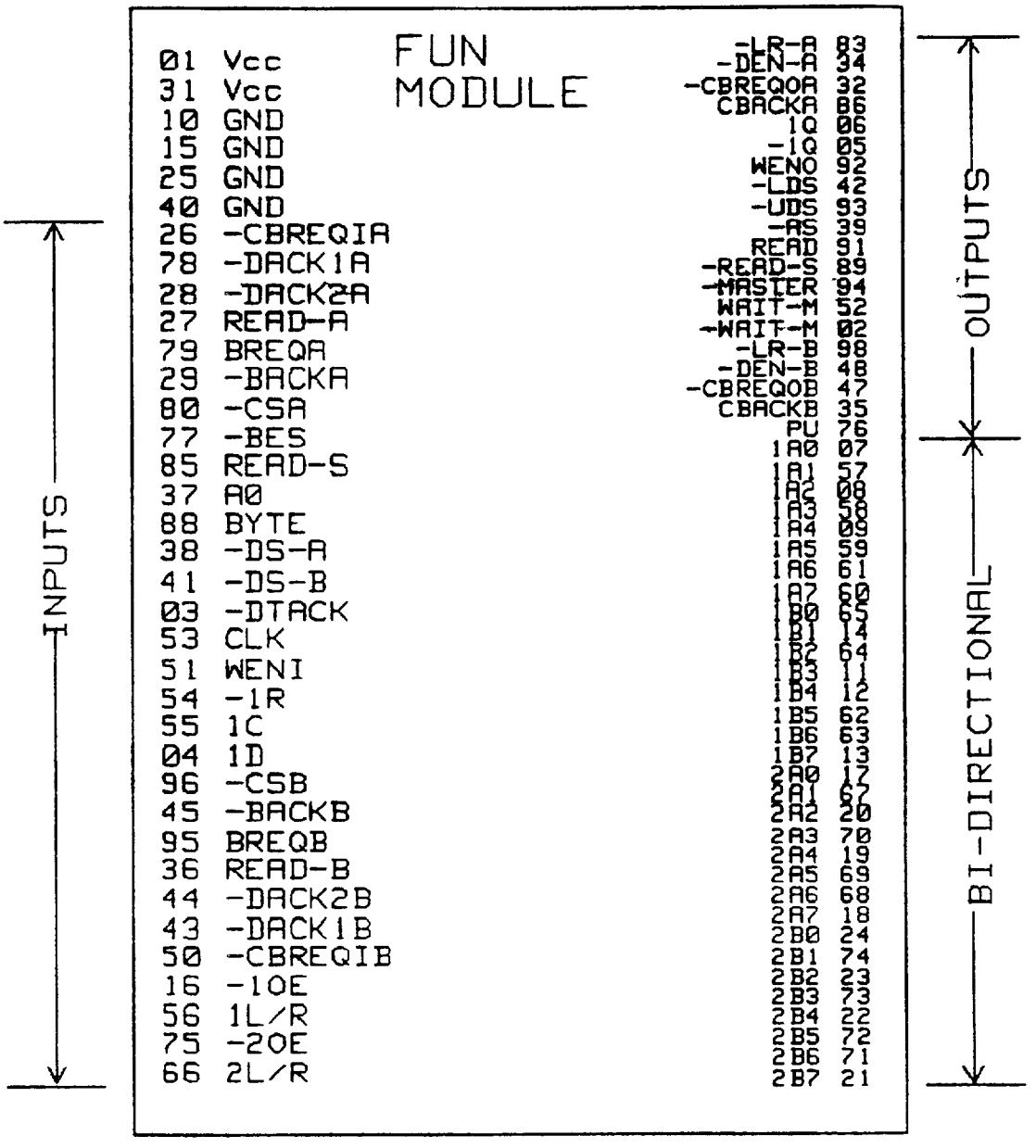


FIGURE 2. FUN pin assignment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

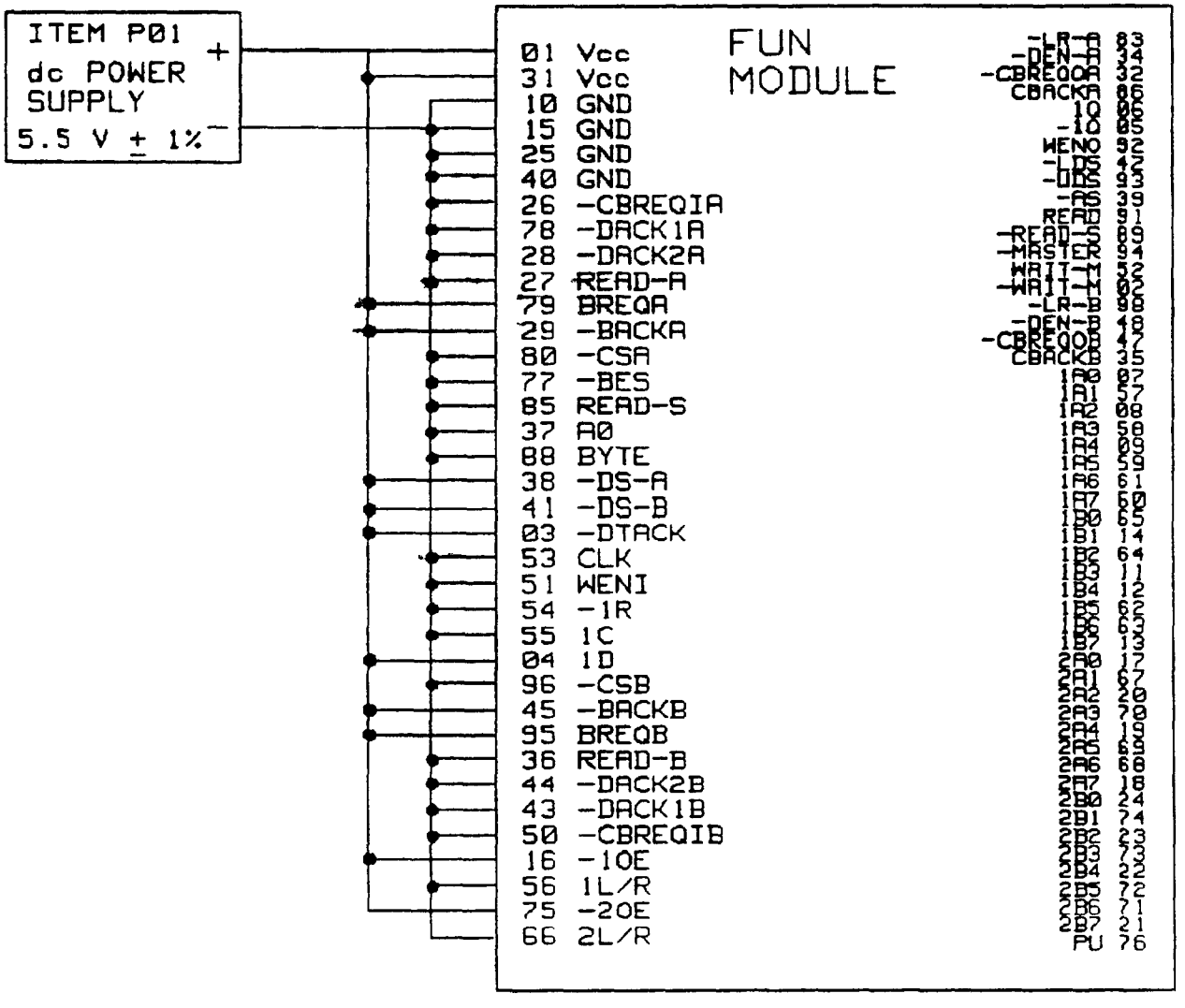
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through V, X and Y shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

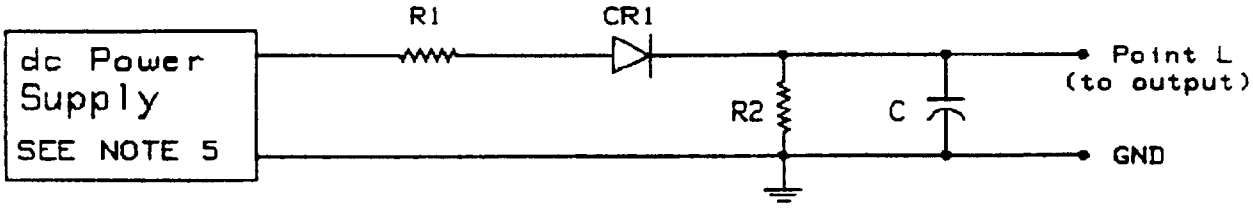


NOTE:

1. The load for each output is as specified on figure 4.

FIGURE 3. FUN operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

FIGURE 4. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

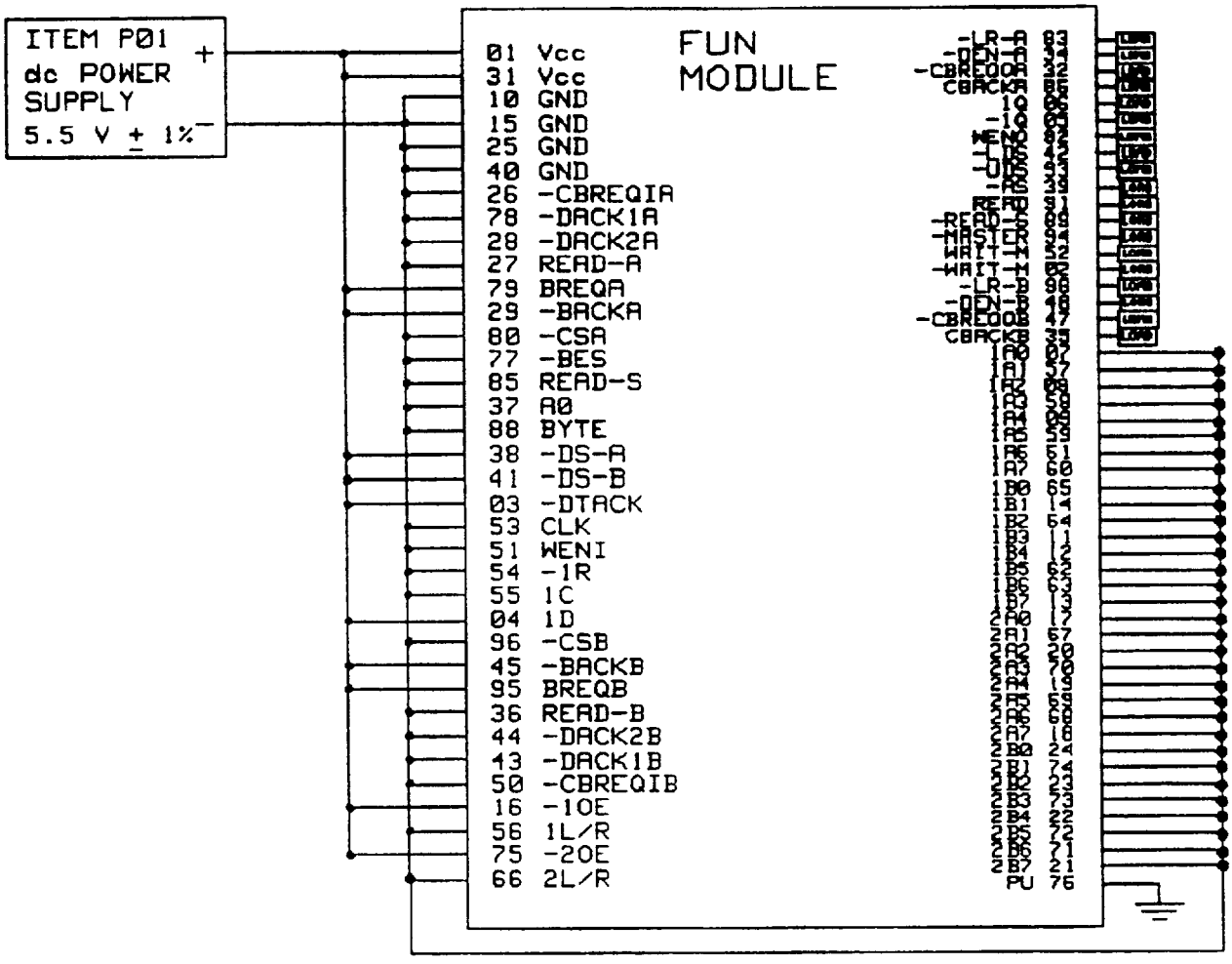
4. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,5,6,52	6250	4.0	0.4
32,34,47,48,83, 98	1200	8.0	0.5
35,86	1215.19	7.75	0.5
42,89,91,92 93,94	1200	12.0	0.5
39	1090.91	11.58	0.5
7,8,9,11,12,13, 14,17,18,19,20, 21,22,23,24,57, 58,59,60,61,62, 63,64,65,67,68, 69,70,71,72,73,74	720.72	11.28	0.4
76	1200	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

5. The load power supply voltage shall be 2.1V ± 1% for disable time test and 2.5V ± 1% for all other tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTE:

1. The load for each output is as specified on figure 4, except for pins 7-9, 11-14, 17-24, 57-65, 67-74, and 76, which are grounded.

FIGURE 5. FUN component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

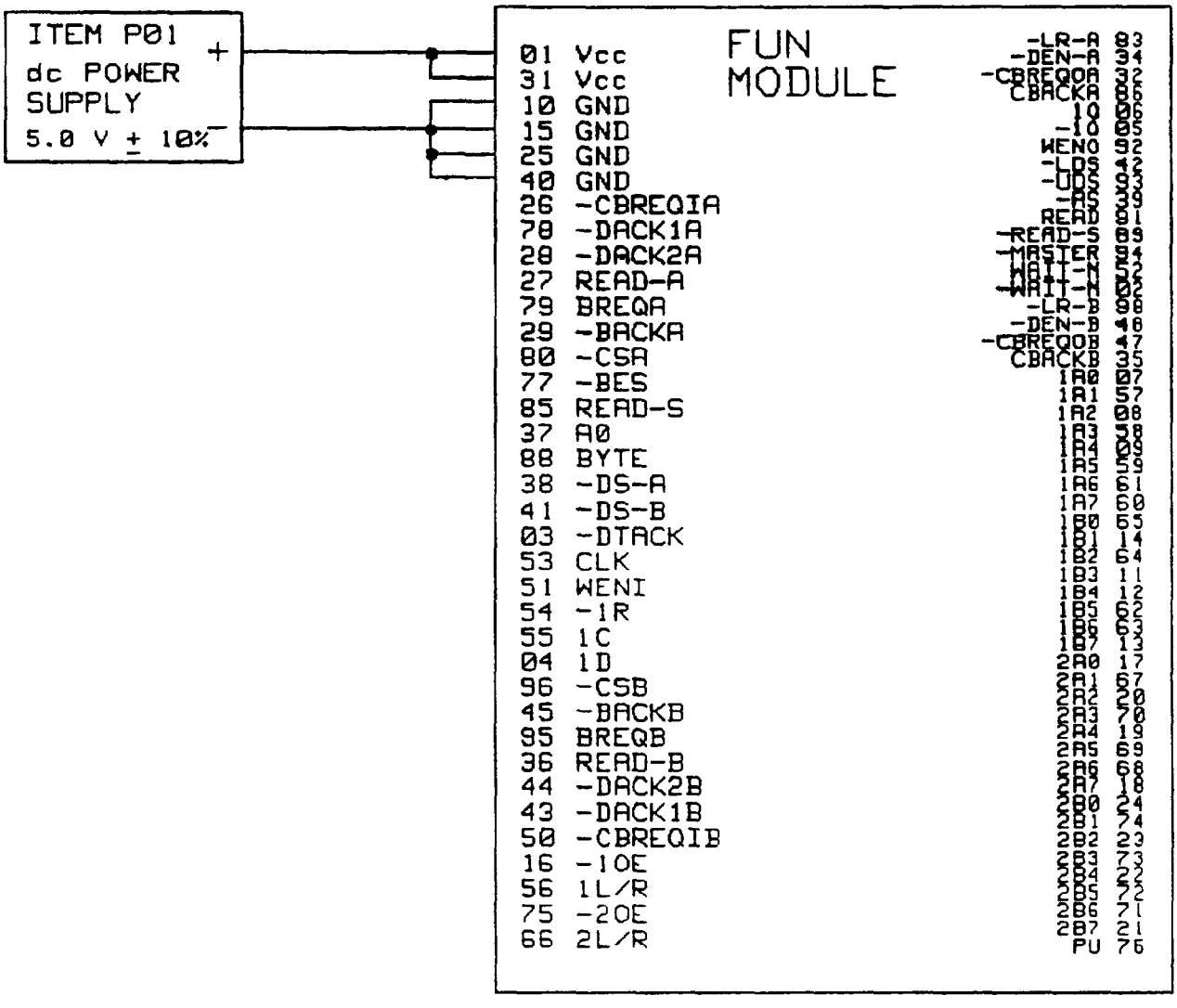


FIGURE 6. FUN vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B, G through V, X, and Y shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through V, X and Y shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.276 W (max); 1.772 W (typical).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters.

Signal	Vic Min (V)	Vil Max (V)	Vih Min (V)	Iil Min (mA)	Iih Max (uA)	Ibd Max (uA)	Cin Max (pF)
-CBREQIA	-1.5	0.8	2.0	-0.25	25.0	1000	55
-CBREQIB	-1.5	0.8	2.0	-0.25	25.0	1000	55
-DACK1A	-1.5	0.8	2.0	-0.25	25.0	1000	55
-DACK2A	-1.5	0.8	2.0	-0.25	25.0	1000	55
-DACK1B	-1.5	0.8	2.0	-2.25	25.0	1000	35
-DACK2B	-1.5	0.8	2.0	-0.25	25.0	1000	25
BREQA	-1.5	0.8	2.0	-0.25	25.0	1000	45
BREQB	-1.5	0.8	2.0	-0.25	25.0	1000	25
-BACKA	-1.5	0.8	2.0	-0.25	25.0	1000	55
-BACKB	-1.5	0.8	2.0	-0.25	25.0	1000	25
-CSA	-1.5	0.8	2.0	-0.25	25.0	1000	25
-CSB	-1.5	0.8	2.0	-0.25	25.0	1000	25
BYTE	-1.5	0.8	2.0	-0.25	25.0	1000	25
A0	-1.5	0.8	2.0	-0.25	25.0	1000	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic Min (V)	Vil Max (V)	Vih Min (V)	Iil Min (mA)	Iih Max (uA)	Ibd Max (uA)	Cin Max (pF)
-DS-A	-1.5	0.8	2.0	-0.25	25.0	1000	25
-DS-B	-1.5	0.8	2.0	-0.25	25.0	1000	35
READ-A	-1.5	0.8	2.0	-0.5	50.0	2000	85
READ-B	-1.5	0.8	2.0	-0.5	50.0	2000	35
-BES	-1.5	0.8	2.0	-0.5	50.0	2000	45
READ-S	-1.5	0.8	2.0	-0.75	75.0	3000	85
-DTACK	-1.5	0.8	2.0	-0.4	20.0	100	35
1D	-1.5	0.7	2.0	-0.4	20.0	100	25
1C	-1.5	0.7	2.0	-0.8	40.0	200	25
CLK	-1.5	0.7	2.0	-0.8	40.0	200	25
WENI	-1.5	0.7	2.0	-0.8	40.0	200	25
-1R	-1.5	0.7	2.0	-1.2	60.0	300	35
1A0	-1.5	0.7	2.0	-1.09	-421.5	100	45
1A1	-1.5	0.7	2.0	-1.09	-421.5	100	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic Min (V)	Vil Max (V)	Vih Min (V)	Iil Min (mA)	Iih Max (uA)	Ibd Max (uA)	Cin Max (pF)
1A2	-1.5	0.7	2.0	-1.09	-421.5	100	45
1A3	-1.5	0.7	2.0	-1.09	-421.5	100	45
1A4	-1.5	0.7	2.0	-1.09	-421.5	100	45
1A5	-1.5	0.7	2.0	-1.09	-421.5	100	45
1A6	-1.5	0.7	2.0	-1.09	-421.5	100	45
1A7	-1.5	0.7	2.0	-1.09	-421.5	100	45
2A0	-1.5	0.7	2.0	-1.09	-421.5	100	45
2A1	-1.5	0.7	2.0	-1.09	-421.5	100	45
2A2	-1.5	0.7	2.0	-1.09	-421.5	100	45
2A3	-1.5	0.7	2.0	-1.09	-421.5	100	45
2A4	-1.5	0.7	2.0	-1.09	-421.5	100	45
2A5	-1.5	0.7	2.0	-1.09	-421.5	100	45
2A6	-1.5	0.7	2.0	-1.09	-421.5	100	45
2A7	-1.5	0.7	2.0	-1.09	-421.5	100	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic Min (V)	Vil Max (V)	Vih Min (V)	Iil Min (mA)	Iih Max (uA)	Ibd Max (uA)	Cin Max (pF)
1B0	-1.5	0.7	2.0	-1.09	-421.5	100	45
1B1	-1.5	0.7	2.0	-1.09	-421.5	100	45
1B2	-1.5	0.7	2.0	-1.09	-421.5	100	45
1B3	-1.5	0.7	2.0	-1.09	-421.5	100	45
1B4	-1.5	0.7	2.0	-1.09	-421.5	100	45
1B5	-1.5	0.7	2.0	-1.09	-421.5	100	45
1B6	-1.5	0.7	2.0	-1.09	-421.5	100	45
1B7	-1.5	0.7	2.0	-1.09	-421.5	100	45
2B0	-1.5	0.7	2.0	-1.09	-421.5	100	45
2B1	-1.5	0.7	2.0	-1.09	-421.5	100	45
2B2	-1.5	0.7	2.0	-1.09	-421.5	100	45
2B3	-1.5	0.7	2.0	-1.09	-421.5	100	45
2B4	-1.5	0.7	2.0	-1.09	-421.5	100	45
2B5	-1.5	0.7	2.0	-1.09	-421.5	100	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.1 Input parameters - Continued.

Signal	Vic Min (V)	Vil Max (V)	Vih Min (V)	Iil Min (mA)	Iih Max (uA)	Ibd Max (uA)	Cin Max (pF)
2B6	-1.5	0.7	2.0	-1.09	-421.5	100	45
2B7	-1.5	0.7	2.0	-1.09	-421.5	100	45
-1OE	-1.5	0.7	2.0	-0.2	20.0	100	45
-2OE	-1.5	0.7	2.0	-0.2	20.0	100	45
1L/R	-1.5	0.7	2.0	-0.2	20.0	100	45
2L/R	-1.5	0.7	2.0	-0.2	20.0	100	45

6.3.2 Output parameters.

Signal	Voh Min (V)	Vol Max (V)	Ioh Min (mA)	Iol Max (mA)	Iozh Min (uA)	Iozl Max (uA)	Ios Min (mA)	Cout Max (pF)
1Q	2.5	0.4	-0.4	4.0	N/A	N/A	-100.0	15
-1Q	2.5	0.4	-0.4	4.0	N/A	N/A	-100.0	15
WAIT-M	2.5	0.4	-0.4	4.0	N/A	N/A	-100.0	15
-WAIT-M	2.5	0.4	-0.4	4.0	N/A	N/A	-100.0	15
1A0	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh Min (V)	Vol Max (V)	Ioh Min (mA)	Iol Max (mA)	Iozh Min (uA)	Iozl Max (uA)	Ios Min (mA)	Cout Max (pF)
1A1	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1A2	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1A3	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1A4	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1A5	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1A6	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1A7	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2A0	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2A1	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2A2	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2A3	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2A4	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2A5	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2A6	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh Min (V)	Vol Max (V)	Ioh Min (mA)	Iol Max (mA)	Iozh Min (uA)	Iozl Max (uA)	Ios Min (mA)	Cout Max (pF)
2A7	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1B0	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1B1	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1B2	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1B3	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1B4	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1B5	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1B6	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
1B7	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2B0	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2B1	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2B2	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2B3	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2B4	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh Min (V)	Vol Max (V)	Ioh Min (mA)	Iol Max (mA)	Iozh Min (uA)	Iozl Max (uA)	Ios Min (mA)	Cout Max (pF)
2B5	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2B6	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
2B7	2.4	0.4	-3.33	11.286	-508.0	-0.804	-225.96	45
READ	2.4	0.5	-2.0	12.0	-100.0	100.0	-250.0	35
-LDS	2.4	0.5	-2.0	12.0	-100.0	100.0	-250.0	35
-UDS	2.4	0.5	-2.0	12.0	-100.0	100.0	-250.0	35
WENO	2.4	0.5	-2.0	12.0	N/A	N/A	-250.0	35
-READ-S	2.4	0.5	-2.0	12.0	N/A	N/A	-250.0	35
-MASTER	2.4	0.5	-2.0	12.0	N/A	N/A	-250.0	45
CBACKA	2.4	0.5	-1.975	7.75	N/A	N/A	-130.27	60
CBACKB	2.4	0.5	-1.975	7.75	N/A	N/A	-130.27	60
-AS	2.4	0.5	-2.2	11.58	-426.3	-386.0	-250.58	35
-LR-A	2.4	0.5	-2.0	8.0	N/A	N/A	-130.0	35
-LR-B	2.4	0.5	-2.0	8.0	N/A	N/A	-130.0	35

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.3.2 Output parameters - Continued.

Signal	Voh Min (V)	Vol Max (V)	Ioh Min (mA)	Iol Max (mA)	Iozh Min (uA)	Iozl Max (uA)	Ios Min (mA)	Cout Max (pF)
-DEN-A	2.4	0.5	-2.0	8.0	N/A	N/A	-130.0	35
-DEN-B	2.4	0.5	-2.0	8.0	N/A	N/A	-130.0	35
-CBREQOA	2.4	0.5	-2.0	8.0	N/A	N/A	-130.0	35
-CBREQOB	2.4	0.5	-2.0	8.0	N/A	N/A	-130.0	35

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The module function is printed in the appendix.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
-DTACK	3	I	This active low input signal indicates that the current DMA bus master data transfer operation may be completed. An active -DTACK input to the FUN causes termination of the -WAIT-M output to the DMA controller module (FUM).
1D 1C -1R	4 55 54	I I I	These signals are the data input, clock input and reset input of a fully pinned out D flip-flop. The preset input of the flip-flop is internally pulled up in the module.
1Q -1Q	6 5	O O	These signals are the Q output and -Q output, respectively, of a fully pinned out D flip-flop. The flip-flop outputs change state after the rising edge of the clock.
WAIT-M	52	O	This active high output signal is the complement of the -WAIT-M output. It is typically not used.
-WAIT-M	2	O	This active low output signal is sent to the DMA controller module (FUM) that is current bus master to allow extension of the DMA transfer operation for slow memory or peripherals. An active low Data Transfer Acknowledge (-DTACK) input to the FUN causes termination of the the -WAIT-M output.
1L/R 2L/R	56 66	I I	These independent input signals control the direction of the two 8-bit three-state transceivers. When 1L/R (2L/R) is high, the transceiver direction is from 1AX to 1BX (2AX to 2BX), and 1BX to 1AX (2BX to 2AX) when 1L/R (2L/R) is low.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
1A0	7	I/O	These signals are the bidirectional Input/Outputs (I/O) of two 8-bit three-state transceivers. The transceivers are typically used to interface the DMA controller module's (FUM) higher order address bits (ADDR17 - ADDR23) and control signals to the rest of the system. Pull-up resistors (6.04 KOhm) are provided on the transceiver I/O so that all signals are constantly defined at a logic high when the transceivers are in a high impedance state.
1A1	57	I/O	
1A2	8	I/O	
1A3	58	I/O	
1A4	9	I/O	
1A5	59	I/O	
1A6	61	I/O	
1A7	60	I/O	
2A0	17	I/O	
2A1	67	I/O	
2A2	20	I/O	
2A3	70	I/O	
2A4	19	I/O	
2A5	69	I/O	
2A6	68	I/O	
2A7	18	I/O	
1B0	65	I/O	
1B1	14	I/O	
1B2	64	I/O	
1B3	11	I/O	
1B4	12	I/O	
1B5	62	I/O	
1B6	63	I/O	
1B7	13	I/O	
2B0	24	I/O	
2B1	74	I/O	
2B2	23	I/O	
2B3	73	I/O	
2B4	22	I/O	
2B5	72	I/O	
2B6	71	I/O	
2B7	21	I/O	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-1OE -2OE	16 75	I I	These independent input signals enable the three-state outputs of the two 8-bit transceivers. When -1OE (-2OE) is active low, the transceiver outputs are enabled. The transceiver outputs are in the high impedance state when -1OE (-2OE) is high.
CLK	53	I	This clock input signal is the same clock used with the two DMA controller modules (FUM) supported by the FUN. It is used in the FUN to synchronize the -WAIT-M output to the DMA controller modules. The -WAIT-M (WAIT-M) outputs change state after the rising edge of the CLK input. The CLK frequency is 4 MHz in a typical system application.
WENI	51	I	This active high input signal is the externally connected Wait Terminate Enable Output (WENO) generated by the FUN module.
-CBREQIA -CBREQIB	26 50	I I	These active low input signals are the externally connected -CBREQOX outputs generated by the FUN module.
-DACK1A -DACK2A -DACK1B -DACK2B	78 28 43 44	I I I I	These active low input signals are sourced by two different DMA controller modules (FUM), where each FUM module provides two channels (A and B) of DMA capability. When active low, these signals indicate that the respective channel of the appropriate DMA module is performing a DMA operation in flyby mode. When a DMA operation in flow-thru mode is being performed, these signals are always high.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
BREQA BREQB	79 95	I I	These active high input signals are sourced by two different DMA control modules (FUM). BREQX is an active high DMA output signal used to request control of the bus from another bus master.
-BACKA -BACKB	29 45	I I	These active low output signals are generated by a Microprocessor Unit (MPU) communications support module (AHH) in response to the FUN Conditioned Bus Request (-CBREQOX) outputs. -BACKX is an active low input indicating that another bus master has relinquished the bus and that no higher priority device has assumed bus control.
-CSA -CSB	80 96	I I	These active low input signals are obtained from the address decoder module (BFM) and are used to select one of the two DMA controller modules (FUM), supported by the FUN module, to be used for a slave read or write operation.
BYTE	88	I	This input signal is supplied by the DMA controller module (FUM) and is used in DMA master operations to select the size of the data transferred. When BYTE is high, a byte (8-bit) transfer is selected. A word (16-bit) transfer is selected when BYTE is low.
A0	37	I	This input signal is supplied by the DMA controller module (FUM) and is used in DMA master operations to condition generation of the FUN module -LDS (A0 high) and -UDS (A0 low) outputs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-DS-A -DS-B	38 41	I I	These active low input signals are sourced by two different DMA controller modules (FUM). A low on -DS-X indicates that the DMA controller is involved in a data transfer operation in the master mode. These signals are used within the FUN module to condition the generation of the -AS, -UDS, and -LDS output signals.
READ-A READ-B	27 36	I I	These active high input signals are sourced by two different DMA controller modules (FUM). The signals indicate that a read of the DMA module is being conducted in either the master or slave modes.
-BES	77	I	This active low input signal is generated by the 68000 support (ASC) module. -BES indicates that a micro-processor bus transfer is in progress. It is distributed throughout the system to enable data buffers.
READ-S	85	I	This active high input signal is typically the read output signal from a microprocessor. In a typical system application, the READ-S input is derived from the read latched (READ LX) outputs of the 68000 support module (ASC).
-AS	39	O	This active low, three-state output signal is supplied to the master control bus and is used to indicate that there is a valid address on the address bus during DMA master operations. -AS is usually transferred to the master control bus by the transceiver logic in the FUN module. A 10 KOhm pull-up resistor is internally connected to the -AS output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
CBACKA CBACKB	86 35	O O	These active high output signals are sent to two different DMA controller modules (FUM) to indicate that the current bus master has relinquished control of the bus and that DMA operations can begin. The CBACKX outputs are a function of the -BACKX inputs conditioned by the -CBREQIX and BREQX inputs as defined in the dynamic parameters.
-LR-A -LR-B	83 98	O O	These output signals are provided to the DMA controller module (FUM) to control the direction of data flow in both the DMA master and slave modes. A DMA read operation is enabled when -LR-X is high and a write when -LR-X is low.
-DEN-A -DEN-B	34 48	O O	These active high output signals are provided to a DMA controller module (FUM) to enable its master data strobe output signal. When -DEN-X is low, the appropriate DMA channel has been selected for a read or write operation (-CSX low), the DMA is in the slave mode (-BACKX high) and data and address buses are enabled (-BES low).
-CBREQOA -CBREQOB	32 47	O O	These active low output signals are provided to a MPU communications support module (AHH) in response to a BREQX input signal to request DMA control of the system bus. The -CBREQOX outputs are externally connected to the -CBREQIX inputs to provide, along with the -BACKX and BREQX signals, the logical conditioning of the -CBREQOX signals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
WENO	92	O	This output goes active high when one of the DMA controller modules (FUM) supported by the FUN is bus master and has supplied an active master Data Strobe (-DS-X) input. The active WENO signal allows the WAIT logic in the FUN to terminate the -WAIT-M signal to the DMA controller module when the -DTACK input to the FUN goes active. WENO is always externally connected to the FUN module Wait Terminate Enable Input (WENI).
-LDS -UDS	42 93	O O	These active low, three-state output signals, in conjunction with the READ output signal, indicate the validity of data on the system data bus when a DMA master operation is being executed. These signals are usually transferred to the master control bus by the transceiver logic in the FUN module.
READ	91	O	This active high, three-state output signal is the read output to the master control bus indicating that a DMA master read operation is required. READ is usually transferred to the master control bus by the transceiver logic in the FUN module.
-READ-S	89	O	This active low output signal is the logical complement of the READ-S input signal.
-MASTER	94	O	This active low output signal indicates that one of the two DMA controller modules (FUM) supported by the FUN module is an active bus master. -MASTER is used to enable the tri-state DMA master outputs -AS, -UDS, -LDS and READ.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
PU	76 .	O	This output is a connection to an internal 1 KOhm pull-up resistor. It is available for general purpose use in tying module inputs to a logic high.

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 4.

(j) Vol test - table XI, figure 4.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 7.

(o) Dynamic test - tables XVI and XVII, figures 8 and 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 10.

(r) Output crosstalk tests - tables XX and XXI, figures 11 and 12.

(s) Output crosstalk duration test - table XXII, figure 13.

(t) Cin test - table XXIII.

(u) Disable time test - table XXIV, figures 14 and 15.

(v) Threshold test - table XXV.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT, AV
Navy - AS, MC
Air Force - 13, 17, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0057)

Agent: NW

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused	30	33,46,49, 81,82,84, 87,90,97, 99,100	10	INF	MOhm	1-7
A4	Isolation, unused to unused	33	46,49,81, 82,87,90, 97,99,100				
A5	Isolation, unused to unused	46	49,81,82, 84,87,90, 97,99,100				
A6	Isolation, unused to unused	49	81,82,84, 87,90,97, 99,10				
A7	Isolation, unused to unused	81	82,84,87, 90,97,99, 100				
A8	Isolation, unused to unused	82	84,87,90, 97,99,100				
A9	Isolation, unused to unused	84	87,90,97, 99,100				
A10	Isolation, unused to unused	87	90,97,99, 100	10	INF	MOhm	1-7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A11	Isolation, unused to unused	90	97,99,100	10	INF	MOhm	1-7
A12	Isolation, unused to unused	97	99,100				
A13	Isolation, unused to unused	99	100	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V 30 V	± 10 percent ± 10 percent	A1 A2-A13
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A13 A1-A13

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity,Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity,GND	10	15				
B3	Continuity,GND	15	25				
B4	Continuity,GND	25	40	0.0	1.0	Ohm	1-4
B5	Continuity,Vcc	1	31	0.0	0.1	Ohm	1-4
B6	Continuity,GND	10	15				
B7	Continuity,GND	15	25				
B8	Continuity,GND	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R10.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.072 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	-DTACK	3	5.5	0.0	0.1	mA	1-3
D2	1D	4					
D3	1A0	7					
D4	1A1	57					
D5	1A2	8					
D6	1A3	58					
D7	1A4	9					
D8	1A5	59					
D9	1A6	61					
D10	1A7	60					
D11	1B0	65					
D12	1B1	14					
D13	1B2	64					
D14	1B3	11					
D15	1B4	12					
D16	1B5	62					
D17	1B6	63					
D18	1B7	13					
D19	2A0	17					
D20	2A1	67	5.5	0.0	0.1	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	2A2	20	5.5	0.0	0.1	mA	1-3
D22	2A3	70					
D23	2A4	19					
D24	2A5	69					
D25	2A6	68					
D26	2A7	18					
D27	2B0	24					
D28	2B1	74					
D29	2B2	23					
D30	2B3	73					
D31	2B4	22					
D32	2B5	72					
D33	2B6	71					
D34	2B7	21					
D35	1L/R	56					
D36	2L/R	66					
D37	-1OE	16					
D38	-2OE	75	5.5	0.0	0.1	mA	1-3
D39	1C	55	5.5	0.0	0.2	mA	1-3
D40	CLK	53	5.5	0.0	0.2	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	WENI	51	5.5	0.0	0.2	mA	1-3
D42	-1R	54	5.5	0.0	0.3	mA	1-3
D43	-CBREQIA	26	5.5	0.0	1.0	mA	1-3
D44	-DACK1A	78					
D45	-DACK2A	28					
D46	BREQA	79					
D47	-BACKA	29					
D48	-CSA	80					
D49	-CBREQIB	50					
D50	-DACK1B	43					
D51	-DACK2B	44					
D52	BREQB	95					
D53	-BACKB	45					
D54	-CSB	96					
D55	BYTE	88					
D56	A0	37					
D57	-DS-A	38					
D58	-DS-B	41	5.5	0.0	1.0	mA	1-3
D59	READ-A	27	5.5	0.0	2.0	mA	1-3
D60	READ-B	36	5.5	0.0	2.0	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D61	-BES	77	5.5	0.0	2.0	mA	1-3
D62	READ-S	85	5.5	0.0	3.0	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V 5.5 V	± 55 mV -55 mV,+0 mV	D1,D2, D35-D62 D3-D34
Power supply current limit	0.894 A	NA	D1-D62
Forcing voltage, (Vin)	5.5 V 5.5 V	± 55 mV -55 mV,+0 mV	D1,D2, D35-D62 D3-D34
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D62 D1-D62
Input current measurement	NA NA NA NA NA NA	± 5 uA ± 10 uA ± 15 uA ± 50 uA ± 100 uA ± 150 uA	D1-D38 D39-D41 D42 D43-D58 D59-D61 D62

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	-DTACK	3	-18.0	-1.5	-0.4	V	1-3
E2	1D	4					
E3	1A0	7					
E4	1A1	57					
E5	1A2	8					
E6	1A3	58					
E7	1A4	9					
E8	1A5	59					
E9	1A6	61					
E10	1A7	60					
E11	1B0	65					
E12	1B1	14					
E13	1B2	64					
E14	1B3	11					
E15	1B4	12					
E16	1B5	62					
E17	1B6	63					
E18	1B7	13					
E19	2A0	17					
E20	2A1	67	-18.0	-1.5	-0.4	V	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	2A2	20	-18.0	-1.5	-0.4	V	1-3
E22	2A3	70					
E23	2A4	19					
E24	2A5	69					
E25	2A6	68					
E26	2A7	18					
E27	2B0	24					
E28	2B1	74					
E29	2B2	23					
E30	2B3	73					
E31	2B4	22					
E32	2B5	72					
E33	2B6	71					
E34	2B7	21					
E35	1L/R	56					
E36	2L/R	66					
E37	-1OE	16					
E38	-2OE	75					
E39	1C	55					
E40	CLK	53	-18.0	-1.5	-0.4	V	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E41	WENI	51	-18.0	-1.5	-0.4	V	1-3
E42	-1R	54					
E43	-CBREQIA	26					
E44	-DACK1A	78					
E45	-DACK2A	28					
E46	BREQA	79					
E47	-BACKA	29					
E48	-CSA	80					
E49	-CBREQIB	50					
E50	-DACK1B	43					
E51	-DACK2B	44					
E52	BREQB	95					
E53	-BACKB	45					
E54	-CSB	96					
E55	BYTE	88					
E56	A0	37					
E57	-DS-A	38					
E58	-DS-B	41					
E59	READ-A	27					
E60	READ-B	36	-18.0	-1.5	-0.4	V	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E61	-BES	77	-18.0	-1.5	-0.4	V	1-3
E62	READ-S	85	-18.0	-1.5	-0.4	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E62
Power supply current limit	0.894 A	NA	E1-E62
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E62
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E62 E1-E62
Input voltage measurement	NA	± 55 mV	E1-E62

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	1Q	6	0.0	-100.0	-15.0	mA	1-4
F2	-1Q	5					
F3	WAIT-M	52					
F4	-WAIT-M	2	0.0	-100.0	-15.0	mA	1-4
F5	1A0	7	0.0	-225.96	-40.87	mA	1-4
F6	1A1	57					
F7	1A2	8					
F8	1A3	58					
F9	1A4	9					
F10	1A5	59					
F11	1A6	61					
F12	1A7	60					
F13	2A0	17					
F14	2A1	67					
F15	2A2	20					
F16	2A3	70					
F17	2A4	19					
F18	2A5	69					
F19	2A6	68					
F20	2A7	18	0.0	-225.96	-40.87	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	1B0	65	0.0	-225.96	-40.87	mA	1-4
F22	1B1	14					
F23	1B2	64					
F24	1B3	11					
F25	1B4	12					
F26	1B5	62					
F27	1B6	63					
F28	1B7	13					
F29	2B0	24					
F30	2B1	74					
F31	2B2	23					
F32	2B3	73					
F33	2B4	22					
F34	2B5	72					
F35	2B6	71					
F36	2B7	21	0.0	-225.96	-40.87	mA	1-4
F37	-LR-A	83	0.0	-130.0	-30.0	mA	1-4
F38	-DEN-A	34					
F39	-CBREQOA	32					
F40	-LR-B	98	0.0	-130.0	-30.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	-DEN-B	48	0.0	-130.0	-30.0	mA	1-4
F42	-CBREQOB	47	0.0	-130.0	-30.0	mA	1-4
F43	WENO	92	0.0	-250.0	-30.0	mA	1-4
F44	-LDS	42					
F45	-UDS	93					
F46	READ	91					
F47	-READ-S	89					
F48	-MASTER	94	0.0	-250.0	-30.0	mA	1-4
F49	CBACKA	86	0.0	-130.27	-30.0	mA	1-4
F50	CBACKB	35	0.0	-130.27	-30.0	mA	1-4
F51	-AS	39	0.0	-250.58	-30.52	mA	1-4

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F51
Power supply current limit	0.894 A	NA	F1-F51
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F51

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	F1-F51
low, (VL)	0.0 V	± 100 mV	F1-F51
Output current measurement	NA	± 4.25 mA	F1-F4
	NA	± 9.25 mA	F5-F36
	NA	± 5.0 mA	F37-F42
	NA	± 11.0 mA	F43-F48
	NA	± 5.01 mA	F49,F50
	NA	± 11.0 mA	F51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	-DTACK	3	2.7	0.0	20.0	uA	1-3
G2	1D	4	2.7	0.0	20.0	uA	1-3
G3	1A0	7	2.7	-508.0	-421.5	uA	1-3
G4	1A1	57					
G5	1A2	8					
G6	1A3	58					
G7	1A4	9					
G8	1A5	59					
G9	1A6	61					
G10	1A7	60					
G11	1B0	65					
G12	1B1	14					
G13	1B2	64					
G14	1B3	11					
G15	1B4	12					
G16	1B5	62					
G17	1B6	63					
G18	1B7	13					
G19	2A0	17					
G20	2A1	67	2.7	-508.0	-421.5	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	2A2	20	2.7	-508.0	-421.5	uA	1-3
G22	2A3	70					
G23	2A4	19					
G24	2A5	69					
G25	2A6	68					
G26	2A7	18					
G27	2B0	24					
G28	2B1	74					
G29	2B2	23					
G30	2B3	73					
G31	2B4	22					
G32	2B5	72					
G33	2B6	71					
G34	2B7	21	2.7	-508.0	-421.5	uA	1-3
G35	1L/R	56	2.7	0.0	20.0	uA	1-3
G36	2L/R	66					
G37	-1OE	16					
G38	-2OE	75	2.7	0.0	20.0	uA	1-3
G39	1C	55	2.7	0.0	40.0	uA	1-3
G40	CLK	53	2.7	0.0	40.0	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G41	WENI	51	2.7	0.0	40.0	uA	1-3
G42	-1R	54	2.7	0.0	60.0	uA	1-3
G43	-CBREQIA	26	2.4	0.0	25.0	uA	1-3
G44	-DACK1A	78					
G45	-DACK2A	28					
G46	BREQA	79					
G47	-BACKA	29					
G48	-CSA	80					
G49	-CBREQIB	50					
G50	-DACK1B	43					
G51	-DACK2B	44					
G52	BREQB	95					
G53	-BACKB	45					
G54	-CSB	96					
G55	BYTE	88					
G56	A0	37					
G57	-DS-A	38					
G58	-DS-B	41	2.4	0.0	25.0	uA	1-3
G59	READ-A	27	2.4	0.0	50.0	uA	1-3
G60	READ-B	36	2.4	0.0	50.0	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G61	-BES	77	2.4	0.0	50.0	uA	1-3
G62	READ-S	85	2.4	0.0	75.0	uA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G62
Power supply current limit	0.894 A	NA	G1-G62
Forcing voltage, (Vih)	2.7 V 2.4 V	± 135 mV ± 120 mV	G1-G42 G43-G62
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	G1-G62 G1-G62
Input current measurement	NA NA NA NA NA NA NA NA	± 1.0 uA ± 4.33 uA ± 1.0 uA ± 2.0 uA ± 3.0 uA ± 1.25 uA ± 2.5 uA ± 3.75 uA	G1,G2 G3-G34 G35-G38 G39-G41 G42 G43-G58 G59-G61 G62

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

(e) All floating inputs should be tied to 0.0 V.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	-DTACK	3	0.4	-0.4	-0.03	mA	1-3
H2	1D	4	0.4	-0.4	-0.03	mA	1-3
H3	1A0	7	0.4	-1.09	-0.804	mA	1-3
H4	1A1	57					
H5	1A2	8					
H6	1A3	58					
H7	1A4	9					
H8	1A5	59					
H9	1A6	61					
H10	1A7	60					
H11	1B0	65					
H12	1B1	14					
H13	1B2	64					
H14	1B3	11					
H15	1B4	12					
H16	1B5	62					
H17	1B6	63					
H18	1B7	13					
H19	2A0	17					
H20	2A1	67	0.4	-1.09	-0.804	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	2A2	20	0.4	-1.09	-0.804	mA	1-3
H22	2A3	70					
H23	2A4	19					
H24	2A5	69					
H25	2A6	68					
H26	2A7	18					
H27	2B0	24					
H28	2B1	74					
H29	2B2	23					
H30	2B3	73					
H31	2B4	22					
H32	2B5	72					
H33	2B6	71					
H34	2B7	21	0.4	-1.09	-0.804	mA	1-3
H35	1L/R	56	0.4	-200.0	0.0	uA	1-3
H36	2L/R	66					
H37	-1OE	16					
H38	-2OE	75	0.4	-200.0	0.0	uA	1-3
H39	1C	55	0.4	-0.8	-0.06	mA	1-3
H40	CLK	53	0.4	-0.8	-0.06	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	WENI	51	0.4	-0.8	-0.06	mA	1-3
H42	-1R	54	0.4	-1.2	-0.06	mA	1-3
H43	-CBREQIA	26	0.4	-0.25	0.0	mA	1-3
H44	-DACK1A	78					
H45	-DACK2A	28					
H46	BREQA	79					
H47	-BACKA	29					
H48	-CSA	80					
H49	-CBREQIB	50					
H50	-DACK1B	43					
H51	-DACK2B	44					
H52	BREQB	95					
H53	-BACKB	45					
H54	-CSB	96					
H55	BYTE	88					
H56	A0	37					
H57	-DS-A	38					
H58	-DS-B	41	0.4	-0.25	0.0	mA	1-3
H59	READ-A	27	0.4	-0.5	0.0	mA	1-3
H60	READ-B	36	0.4	-0.5	0.0	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H61	-BES	77	0.4	-0.5	0.0	mA	1-3
H62	READ-S	85	0.4	-0.75	0.0	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H62
Power supply current limit	0.894 A	NA	H1-H62
Forcing voltage, (Vil)	0.4 V	± 20 mV	H1-H62
Pattern voltage high, (VH)	3.0 V	± 100 mV	H1-H62
low, (VL)	0.0 V	± 100 mV	H1-H62
Input current measurement	NA	± 18.5 uA	H1,H2
	NA	± 14.3 uA	H3-H34
	NA	± 10.0 uA	H35-H38
	NA	± 37.0 uA	H39-H41
	NA	± 57.0 uA	H42
	NA	± 12.5 uA	H43-H58
	NA	± 25.0 uA	H59-H61
	NA	± 37.5 uA	H62

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	1Q	6	-0.4	2.5	4.5	V	1-4
J2	-1Q	5					
J3	WAIT-M	52					
J4	-WAIT-M	2	-0.4	2.5	4.5	V	1-4
J5	1A0	7	-3.33	2.4	4.5	V	1-4
J6	1A1	57					
J7	1A2	8					
J8	1A3	58					
J9	1A4	9					
J10	1A5	59					
J11	1A6	61					
J12	1A7	60					
J13	2A0	17					
J14	2A1	67					
J15	2A2	20					
J16	2A3	70					
J17	2A4	19					
J18	2A5	69					
J19	2A6	68					
J20	2A7	18	-3.33	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	1B0	65	-3.33	2.4	4.5	V	1-4
J22	1B1	14					
J23	1B2	64					
J24	1B3	11					
J25	1B4	12					
J26	1B5	62					
J27	1B6	63					
J28	1B7	13					
J29	2B0	24					
J30	2B1	74					
J31	2B2	23					
J32	2B3	73					
J33	2B4	22					
J34	2B5	72					
J35	2B6	71					
J36	2B7	21	-3.33	2.4	4.5	V	1-4
J37	-AS	39	-2.2	2.4	4.5	V	1-4
J38	CBACKA	86	-1.975	2.4	4.5	V	1-4
J39	CBACKB	35	-1.975	2.4	4.5	V	1-4
J40	-LR-A	83	-2.0	2.4	4.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	-LR-B	98	-2.0	2.4	4.5	V	1-4
J42	-DEN-A	34					
J43	-DEN-B	48					
J44	-CBREQOA	32					
J45	-CBREQOB	47					
J46	WENO	92					
J47	-LDS	42					
J48	-UDS	93					
J49	READ	91					
J50	-READ-S	89					
J51	-MASTER	94	-2.0	2.4	4.5	V	1-4
J52	PU	76	-2.0	2.4	2.6	V	1-4

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J52
Power supply current limit	0.894 A	NA	J1-J52

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Ioh)	-0.4 mA -3.33 mA -2.2 mA -1.975 mA -2.0 mA	± 4.0 uA ± 33.3 uA ± 22.0 uA ± 19.75 uA ± 20.0 uA	J1-J4 J5-J36 J37 J38,J39 J40-J52
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J52 J1-J52
Output voltage measurement	NA NA NA	± 100 mV ± 105 mV ± 10 mV	J1-J4 J5-J51 J52

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 4.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K1	1Q	6	4.0	0.0	0.4	V	1-4
K2	-1Q	5					
K3	WAIT-M	52					
K4	-WAIT-M	2	4.0	0.0	0.4	V	1-4
K5	1A0	7	11.286	0.0	0.4	V	1-4
K6	1A1	57					
K7	1A2	8					
K8	1A3	58					
K9	1A4	9					
K10	1A5	59					
K11	1A6	61					
K12	1A7	60					
K13	2A0	17					
K14	2A1	67					
K15	2A2	20					
K16	2A3	70					
K17	2A4	19					
K18	2A5	69					
K19	2A6	68					
K20	2A7	18	11.286	0.0	0.4	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	1B0	65	11.286	0.0	0.4	V	1-4
K22	1B1	14					
K23	1B2	64					
K24	1B3	11					
K25	1B4	12					
K26	1B5	62					
K27	1B6	63					
K28	1B7	13					
K29	2B0	24					
K30	2B1	74					
K31	2B2	23					
K32	2B3	73					
K33	2B4	22					
K34	2B5	72					
K35	2B6	71					
K36	2B7	21	11.286	0.0	0.4	V	1-4
K37	-AS	39	11.58	0.0	0.5	V	1-4
K38	CBACKA	86	7.75	0.0	0.5	V	1-4
K39	CBACKB	35	7.75	0.0	0.5	V	1-4
K40	-LR-A	83	8.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K41	-LR-B	98	8.0	0.0	0.5	V	1-4
K42	-DEN-A	34					
K43	-DEN-B	48					
K44	-CBREQOA	32					
K45	-CBREQOB	47	8.0	0.0	0.5	V	1-4
K46	WENO	92	12.0	0.0	0.5	V	1-4
K47	-LDS	42					
K48	-UDS	93					
K49	READ	91					
K50	-READ-S	89					
K51	-MASTER	94	12.0	0.0	0.5	V	1-4

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K51
Power supply current limit	0.894 A	NA	K1-K51

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Forcing current, (Iol)	4.0 uA	± 40.0 uA	K1-K4
	11.286 uA	± 112.86 uA	K5-K36
	11.58 uA	± 115.8 uA	K37
	7.75 uA	± 77.5 uA	K38,K39
	8.0 uA	± 80.0 uA	K40-K45
	12.0 uA	± 120.0 uA	K46-K51
Pattern voltage high, (VH)	3.0 V	± 100 mV	K1-K51
low, (VL)	0.0 V	± 100 mV	K1-K51
Output voltage measurement	NA	± 20 mV	K1-K36
	NA	± 25 mV	K37-K51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 4.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	1A0	7	2.7	-508.0	-421.5	uA	1-3
L2	1A1	57					
L3	1A2	8					
L4	1A3	58					
L5	1A4	9					
L6	1A5	59					
L7	1A6	61					
L8	1A7	60					
L9	2A0	17					
L10	2A1	67					
L11	2A2	20					
L12	2A3	70					
L13	2A4	19					
L14	2A5	69					
L15	2A6	68					
L16	2A7	18					
L17	1B0	65					
L18	1B1	14					
L19	1B2	64					
L20	1B3	11	2.7	-508.0	-421.5	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	1B4	12	2.7	-508.0	-421.5	uA	1-3
L22	1B5	62					
L23	1B6	63					
L24	1B7	13					
L25	2B0	24					
L26	2B1	74					
L27	2B2	23					
L28	2B3	73					
L29	2B4	22					
L30	2B5	72					
L31	2B6	71					
L32	2B7	21	2.7	-508.0	-421.5	uA	1-3
L33	-LDS	42	2.4	-100.0	100.0	uA	1-3
L34	-UDS	93					
L35	READ	91	2.4	-100.0	100.0	uA	1-3
L36	-AS	39	2.4	-426.3	-195.2	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L36
Power supply current limit	0.894 A	NA	L1-L36
Forcing voltage, (Voh)	2.7 V 2.4 V	± 135 mV ± 120 mV	L1-L32 L33-L36
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L36 L1-L36
Output current measurement	NA NA NA	± 4.33 uA ± 10.0 uA ± 11.6 uA	L1-L32 L33-L35 L36

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	1A0	7	0.4	-1.09	-0.804	uA	1-3
M2	1A1	57					
M3	1A2	8					
M4	1A3	58					
M5	1A4	9					
M6	1A5	59					
M7	1A6	61					
M8	1A7	60					
M9	2A0	17					
M10	2A1	67					
M11	2A2	20					
M12	2A3	70					
M13	2A4	19					
M14	2A5	69					
M15	2A6	68					
M16	2A7	18					
M17	1B0	65					
M18	1B1	14					
M19	1B2	64					
M20	1B3	11	0.4	-1.09	-0.804	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M21	1B4	12	0.4	-1.09	-0.804	uA	1-3
M22	1B5	62					
M23	1B6	63					
M24	1B7	13					
M25	2B0	24					
M26	2B1	74					
M27	2B2	23					
M28	2B3	73					
M29	2B4	22					
M30	2B5	72					
M31	2B6	71					
M32	2B7	21	0.4	-1.09	-0.804	uA	1-3
M33	-LDS	42	0.4	-100.0	100.0	uA	1-3
M34	-UDS	93					
M35	READ	91	0.4	-100.0	100.0	uA	1-3
M36	-AS	39	0.4	-637.0	-386.0	uA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M36
Power supply current limit	0.894 A	NA	M1-M36
Forcing voltage, (Vol)	0.4 V	± 20 mV	M1-M36
Pattern voltage high, (VH)	3.0 V	± 100 mV	M1-M36
low, (VL)	0.0 V	± 100 mV	M1-M36
Output current measurement	NA	± 14.3 nA	M1-M32
	NA	± 10.0 uA	M33-M35
	NA	± 12.6 uA	M36

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	59	596	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	0.894 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 26.8 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.894 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 20 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 100 mV ± 150 mV ± 20 mV	 P1 P2 P1,P2

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

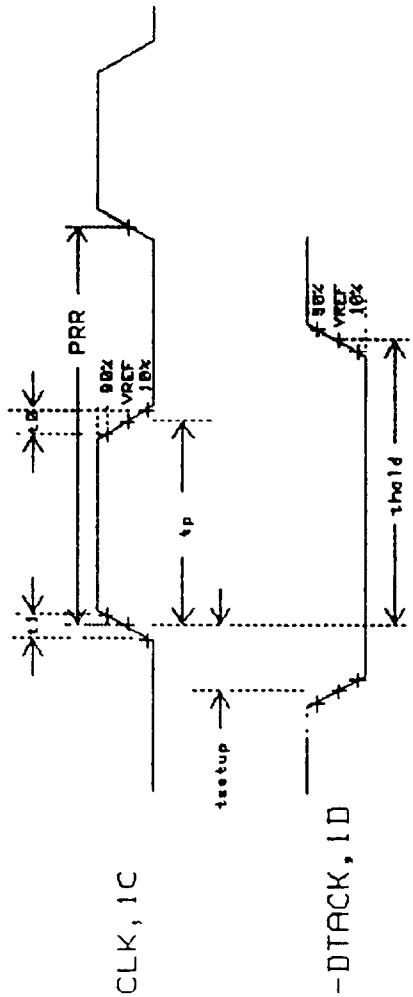
(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 4). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 7.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input parameters:

   PRR = 50 nS minimum.
   tP = 25 nS (both twh and twl), minimum.
   tsetup = 20 nS.
   thold = 50 nS.
   t1 = 15 nS.
   t0 = 6 nS.

2. VREF = 1.3 ± 1.0 V.

3. Pattern voltages shall be as specified in the test tolerance table.

FIGURE 7. Input timing parameters for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-7
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-7
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-7
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-7

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	0.894 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 150 mV ± 100 mV	 Q1-Q4 Q1-Q4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 150 mV	Q2,Q4
low, (Vol)	NA	± 20 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

3. The load circuit for this test is specified on figure 4.

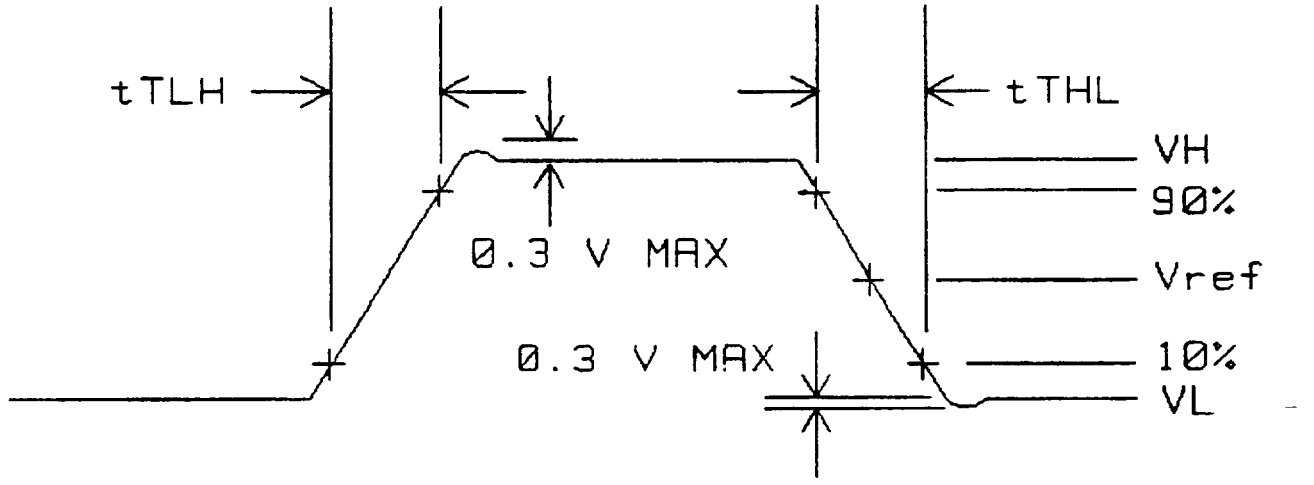
4. All of the input waveforms shall meet the requirements shown on figure 8.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. I/O waveforms are as referenced on figure 9.

7. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
3,4,7,57,8,58,9,59,61,60,65, 69,14,64,11,12,62,63,13,17, 67,20,70,19,68,18,24,74,23, 73,22,72,71,21,56,66,16,75, 55,53,51,54,26,78,28,79,29, 80,50,43,44,95,45,96,88,37, 38,41,27,36,77,85	15 ± 1.5 nS	6 ± 1.5 nS

FIGURE 8. Sample input waveform for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
3	1-8	1A0-1A7 to 1B0-1B7; tPLH	-	-	-	28	23	28	nS	9a
3	9-16	2A0-2A7 to 2B0-2B7; tPLH	-	-	-	28	23	28	nS	9a
3	17-24	1B0-1B7 to 1A0-1A7; tPLH	-	-	-	28	23	28	nS	9a
3	25-32	2B0-2B7 to 2A0-2A7; tPLH	-	-	-	28	23	28	nS	9a
4	33-40	1A0-1A7 to 1B0-1B7; tPHL	-	-	-	30	25	30	nS	9a
4	41-48	2A0-2A7 to 2B0-2B7; tPHL	-	-	-	30	25	30	nS	9a
4	49-56	1B0-1B7 to 1A0-1A7; tPHL	-	-	-	30	25	30	nS	9a
4	57-64	2B0-2B7 to 2A0-2A7; tPHL	-	-	-	30	25	30	nS	9a
11	65-96	1A0-1A7, 2A0-2A7, 1B0-1B7, 2B0-2B7; tPZH	-	-	-	64	51	64	nS	9b
12	97-128	1A0-1A7, 2A0-2A7, 1B0-1B7, 2B0-2B7; tPZL	-	-	-	66	53	66	nS	9b
1	129-131	-BACKB, -BES, -CSB to -DEN-B; tPLH	-	-	-	52	52	52	nS	9a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
2	132-134	-BACKB, -BES, -CSB to -DEN-B; tPHL	-	-	-	56	56	56	nS	9a
1	135-140	READ-S, -DACK1B, -DACK2B, -CSB, READ-B, -BACKB to -LR-B; tPLH	-	-	-	52	52	52	nS	9a
2	141-146	READ-S, -DACK1B, -DACK2B, -CBS, READ-B, -BACKB to -LR-B; tPHL	-	-	-	56	56	56	nS	9a
1	147-149	-BACKB, -CBREQIB, BREQB to -CBREQOB; tPLH	-	-	-	52	52	52	nS	9a
2	150-152	-BACKB, -CBREQIB, BREQB to -CBREQOB; tPHL	-	-	-	56	56	56	nS	9a
1	153-155	BREQB, -CBREQIB, -BACKB to CBACKB; tPLH	-	-	-	52	52	52	nS	9a
2	156-158	BREQB, -CBREQIB, -BACKB to CBACKB; tPHL	-	-	-	56	56	56	nS	9a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
1	159-161	-BACKA, -BES, -CSA to -DEN-A; tPLH	-	-	-	52	52	52	nS	9a
2	162-164	-BACKA, -BES, -CSA to -DEN-A; tPHL	-	-	-	56	56	56	nS	9a
1	165-170	READ-S, -DACK1A, -DACK2A, -CSA, READ-A, -BACKA to -LR-A; tPLH	-	-	-	52	52	52	nS	9a
2	171-176	READ-S, -DACK1A, -DACK2A, -CSA, READ-A, -BACKA to -LR-A; tPHL	-	-	-	56	56	56	nS	9a
1	177-179	-BACKA, -CBREQIA, BREQA to -CBREQOA; tPLH	-	-	-	52	52	52	nS	9a
2	180-182	-BACKA, -CBREQIA, BREQA to -CBREQOA; tPHL	-	-	-	56	56	56	nS	9a
1	183-185	BREQA, -CBREQIA, -BACKA to CBACKA; tPLH	-	-	-	52	52	52	nS	9a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
2	186-188	BREQA, -CBREQIA, -BACKA to CBACKA; tPHL	-	-	-	56	56	56	nS	9a
1	189	READ-S to -READ-S; tPLH	-	-	-	37	37	37	nS	9a
2	190	READ-S to -READ-S; tPHL	-	-	-	38	38	38	nS	9a
1	191-198	-DS-A, -DS-B to WENO, -AS, -UDS, -LDS; tPLH	-	-	-	37	37	37	nS	9a
2	199-206	-DS-A, -DS-B to WENO, -AS, -UDS, -LDS; tPHL	-	-	-	38	38	38	nS	9a
1	207-210	A0, BYTE to -UDS, -LDS; tPLH	-	-	-	37	37	37	nS	9a
2	211-214	A0, BYTE to -UDS, -LDS; tPHL	-	-	-	38	38	38	nS	9a
1	215-244	BREQB, BREQA, -CBREQIB, -BACKB, -CBREQIA, -BACKA to -MASTER, -AS, -UDS, -UDS, -LDS, READ; tPLH	-	-	-	89	89	89	nS	9a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
2	245-274	BREQB, BREQA, -CBREQIB, -BACKB, -CBREQIA, -BACKA to -MASTER, -AS, -UDS, -LDS, READ; tPHL	-	-	-	94	94	94	nS	9a
1	275-276	READ-A, READ-B to READ; tPLH	-	-	-	37	37	37	nS	9a
2	277-278	READ-A, READ-B to READ; tPHL	-	-	-	38	38	38	nS	9a
17	279	-1R to -1Q; tPLH	5	5	5	51	42	51	nS	9b
17	280	-1R to 1Q; tPHL	5	5	5	70	57	70	nS	9b
17	281	WENI to WAIT-M; tPLH	5	5	5	51	42	51	nS	9b
17	282	WENI to -WAIT-M, tPHL	5	5	5	70	57	70	nS	9b
9, 16	283-284	1C to 1Q, -1Q; tPLH	5	5	5	51	42	51	nS	9a
10, 15	285-286	1C to 1Q, -1Q; tPHL	5	5	5	70	57	70	nS	9a
9, 16	287-288	CLK to WAIT-M, -WAIT-M; tPLH	5	5	5	51	42	51	nS	9a
10, 15	289-290	CLK to WAIT-M, -WAIT-M; tPHL	5	5	5	70	57	70	nS	9a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
13	291-314	BREQB, BREQA, -CBREQIB, -CBREQIA, -BACKB, -BACKA to -AS, -UDS, -LDS, READ; tPZL	-	-	-	94	94	94	nS	9b
14	315-338	BREQB, BREQA, -CBREQIB, -CBREQIA, -BACKB, -BACKA to -AS, -UDS, -LDS, READ; tPZH	-	-	-	89	89	89	nS	9b
5	339	1C Maximum Frequency	10	10	10	-	-	-	MHz	9a
5	340	CLK maximum frequency	10	10	10	-	-	-	MHz	9a
8	341-342	WENI, -DTACK setup time to CLK; tSU	20	20	20	-	-	-	nS	9a
7	343	-DTACK hold time to CLK; tH	5	5	5	-	-	-	nS	9a
8	344-345	-1R, 1D setup time to 1C; tSU	20	20	20	-	-	-	nS	9a
7	346	1D hold time to 1C; tH	5	5	5	-	-	-	nS	9a
6	347-348	CLK, 1C pulse width high; PWH	25	25	25	-	-	-	nS	9a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
6	349-350	WENI, -1R pulse width low; PWL	25	25	25	-	-	-	nS	9a
3	351-358	1A0-1A7 to 1B0-1B7; tPLH	-	-	-	37	32	37	nS	9a
3	359-366	2A0-2A7 to 2B0-2B7; tPLH	-	-	-	37	32	37	nS	9a
3	367-374	1B0-1B7 to 1A0-1A7; tPLH	-	-	-	37	32	37	nS	9a
3	375-382	2B0-2B7 to 2A0-2A7; tPLH	-	-	-	37	32	37	nS	9a
4	383-390	1A0-1A7 to 1B0-1B7; tPHL	-	-	-	41	36	41	nS	9a
4	391-398	2A0-2A7 to 2B0-2B7; tPHL	-	-	-	41	36	41	nS	9a
4	399-406	1B0-1B7 to 1A0-1A7; tPHL	-	-	-	41	36	41	nS	9a
4	407-414	2B0-2B7 to 2A0-2A7; tPHL	-	-	-	41	36	41	nS	9a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

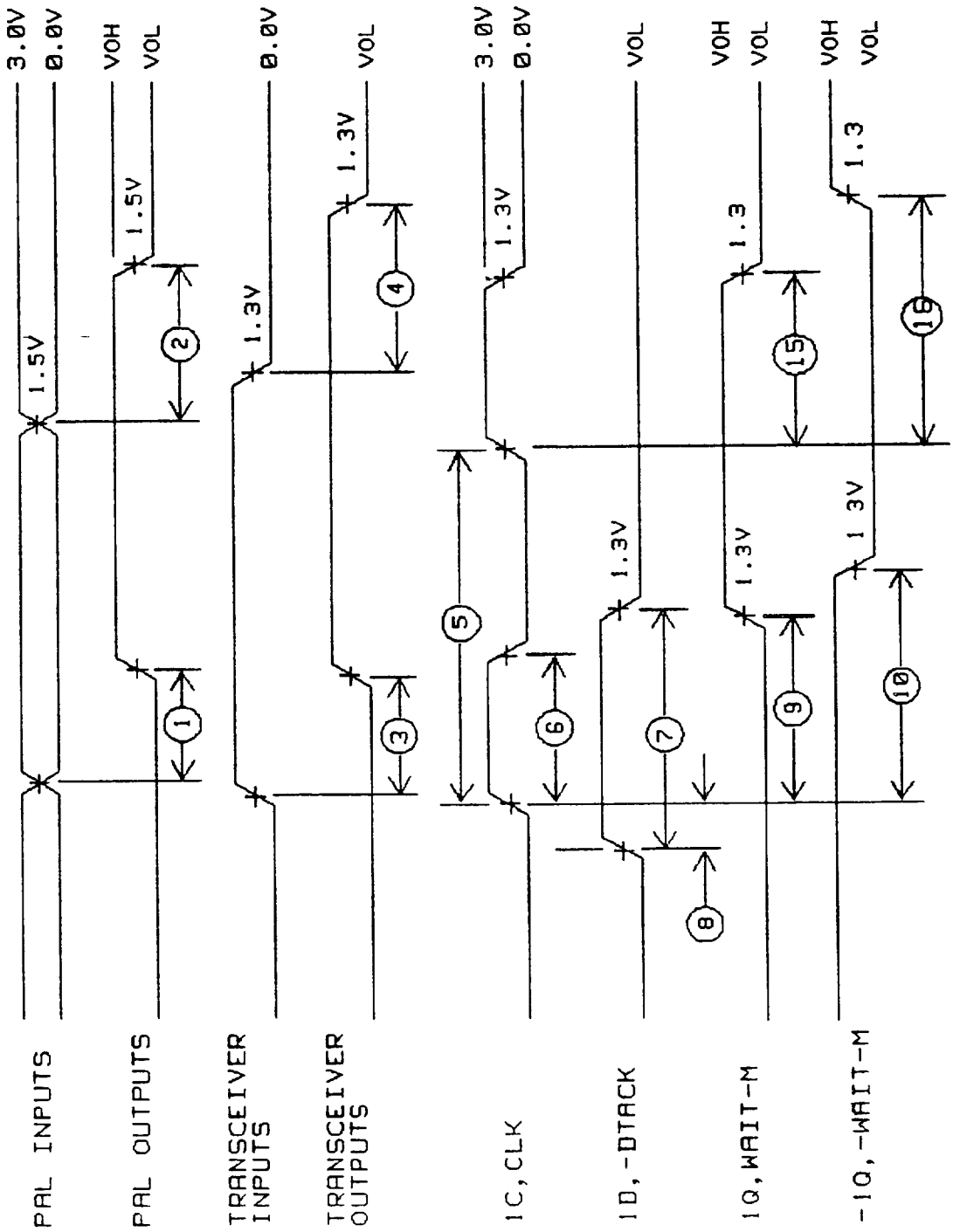


FIGURE 9a. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

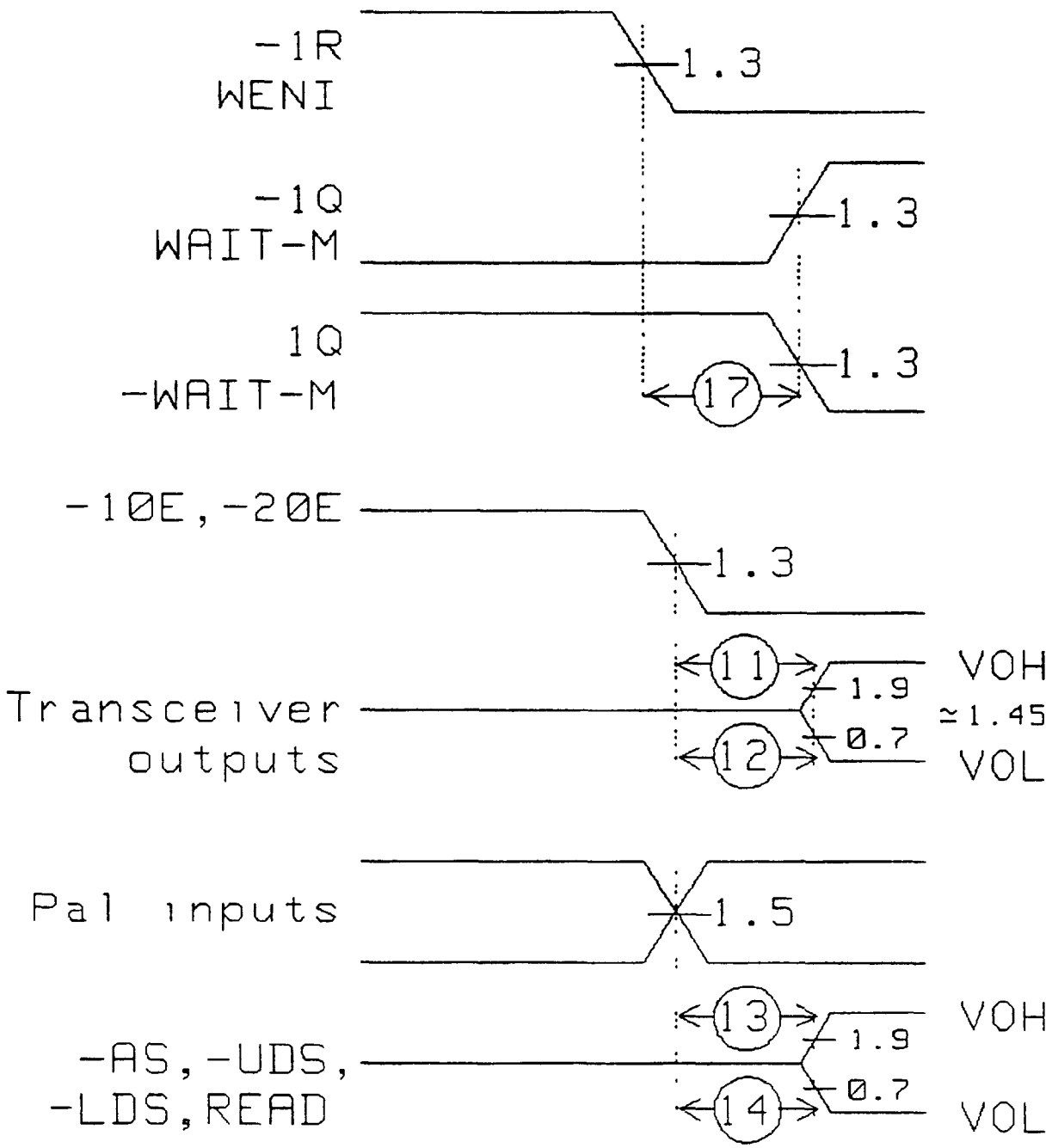


FIGURE 9b. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.7	6.9	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 160 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1,31	0	400	mV	1-3

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	0.894 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D07, inspect for the most negative spike as shown on figure 10 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 12 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

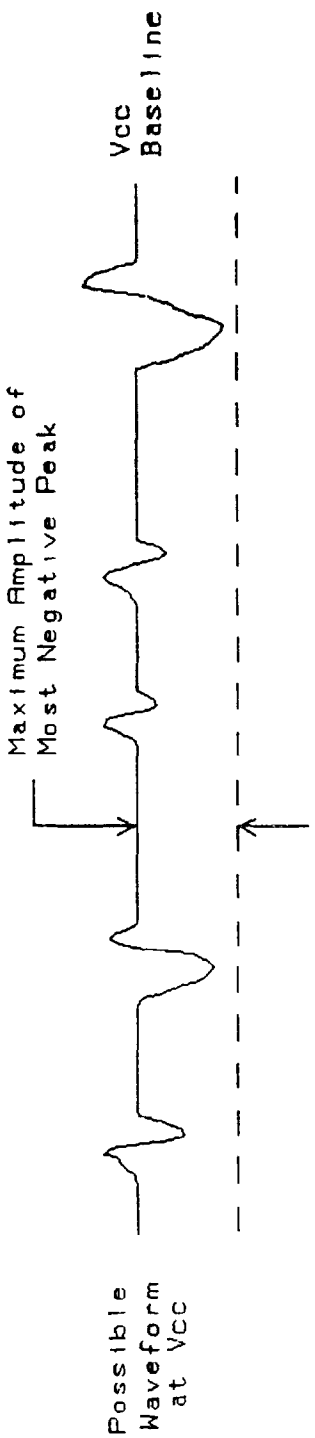


FIGURE 10. Power supply transient amplitude waveform for table XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T1	-LR-A	83	0	500	mV	1-4	11,12
T2	-DEN-A	34					
T3	-CBREQOA	32					
T4	CBACKA	86					
T5	-LR-B	98					
T6	-DEN-B	48					
T7	-CBREQOB	47					
T8	CBACKB	35					
T9	WENO	92					
T10	-LDS	42					
T11	-UDS	93					
T12	-AS	39					
T13	READ	91					
T14	-READ-S	89					
T15	-MASTER	94	0	500	mV	1-4	11,12
T16	1Q	6	0	400	mV	1-4	11,12
T17	-1Q	5					
T18	WAIT-M	52					
T19	-WAIT-M	2	0	400	mV	1-4	11,12
T20	1A0	7	0	525	mV	1-4	11,12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T21	1A1	57	0	525	mV	1-4	11,12
T22	1A2	8					
T23	1A3	58					
T24	1A4	9					
T25	1A5	59					
T26	1A6	61					
T27	1A7	60					
T28	2A0	17					
T29	2A1	67					
T30	2A2	20					
T31	2A3	70					
T32	2A4	19					
T33	2A5	69					
T34	2A6	68					
T35	2A7	18					
T36	1B0	65					
T37	1B1	14					
T38	1B2	64					
T39	1B3	11					
T40	1B4	12	0	525	mV	1-4	11,12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T41	1B5	62	0	525	mV	1-4	11,12
T42	1B6	63					
T43	1B7	13					
T44	2B0	24					
T45	2B1	74					
T46	2B2	23					
T47	2B3	73					
T48	2B4	22					
T49	2B5	72					
T50	2B6	71					
T51	2B7	21	0	525	mV	1-4	11,12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U1	-LR-A	83	0	500	mV	1-4	11,12
U2	-DEN-A	34					
U3	-CBREQOA	32					
U4	CBACKA	86					
U5	-LR-B	98					
U6	-DEN-B	48					
U7	-CBREQOB	47					
U8	CBACKB	35					
U9	WENO	92					
U10	-LDS	42					
U11	-UDS	93					
U12	-AS	39	0	500	mV	1-4	11,12
U13	READ	91	0	800	mV	1-4	11,12
U14	-READ-S	89	0	500	mV	1-4	11,12
U15	-MASTER	94	0	800	mV	1-4	11,12
U16	1Q	6	0	400	mV	1-4	11,12
U17	-1Q	5					
U18	WAIT-M	52					
U19	-WAIT-M	2	0	400	mV	1-4	11,12
U20	1A0	7	0	925	mV	1-4	11,12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U21	1A1	57	0	925	mV	1-4	11,12
U22	1A2	8					
U23	1A3	58					
U24	1A4	9					
U25	1A5	59					
U26	1A6	61					
U27	1A7	60					
U28	2A0	17					
U29	2A1	67					
U30	2A2	20					
U31	2A3	70					
U32	2A4	19					
U33	2A5	69					
U34	2A6	68					
U35	2A7	18					
U36	1B0	65					
U37	1B1	14					
U38	1B2	64					
U39	1B3	11					
U40	1B4	12	0	925	mV	1-4	11,12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
U41	1B5	62	0	925	mV	1-4	11,12
U42	1B6	63					
U43	1B7	13					
U44	2B0	24					
U45	2B1	74					
U46	2B2	23					
U47	2B3	73					
U48	2B4	22					
U49	2B5	72					
U50	2B6	71					
U51	2B7	21	0	925	mV	1-4	11,12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

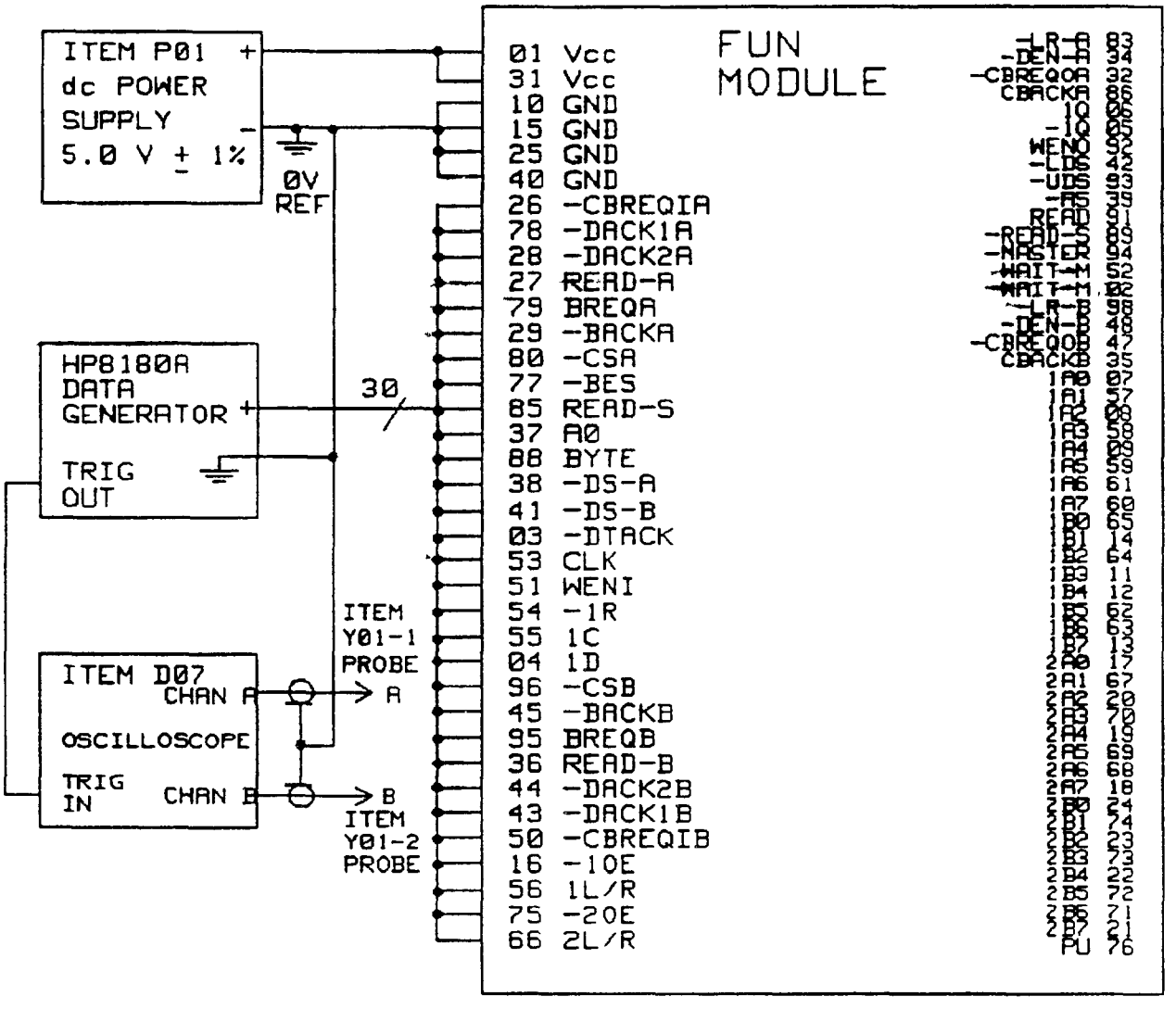


FIGURE 11. Output crosstalk, high and low level, test circuit for tables XX and XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

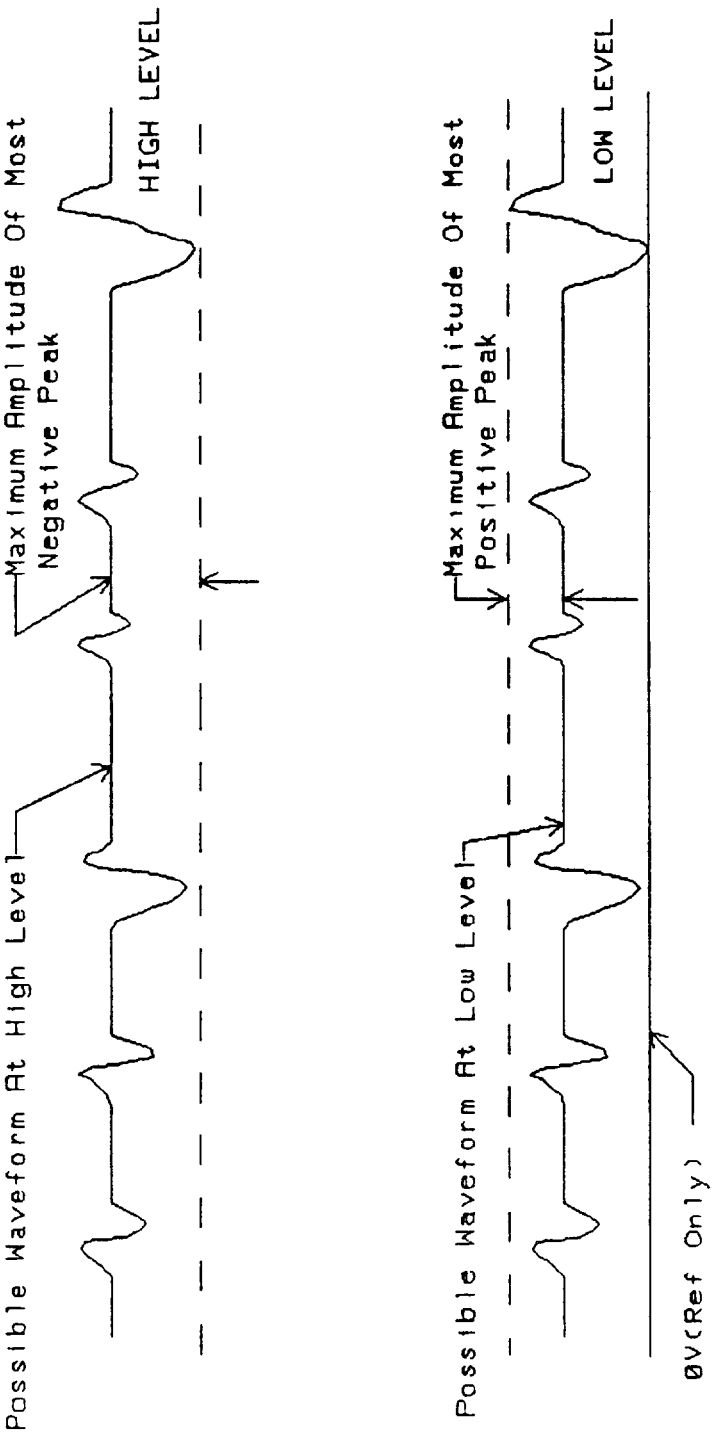


FIGURE 12. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output crosstalk (high and low level) test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply, (Vcc)	5.0 V 5.0 V	± .25 mV ± .25 mV	894 mA 894 mA	T1-T51 U1-U51
Pattern voltage high, (VH) high, (VH) low, (VL) low, (VL)	 3.0 V 3.0 V 0.0 V 0.0 V	 ± 100 mV ± 100 mV ± 100 mV ± 100 mV	 NA NA NA NA	 T1-T51 U1-U51 T1-T51 U1-U51
Output crosstalk measurement	NA NA NA NA NA NA NA	± 25 mV ± 25 mV ± 40 mV ± 20 mV ± 20 mV ± 26.25 mV ± 46.25 mV	NA NA NA NA NA NA NA	T1-T15 U1-U12,U14 U13,U15 T16-T19 U16-U19 T20-T51 U20-U51

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in tables XX and XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 5.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXII. Output crosstalk, duration, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
V1	1A0	7	0	61	nS	1-4	13
V2	1A1	57					
V3	1A2	8					
V4	1A3	58					
V5	1A4	9					
V6	1A5	59					
V7	1A6	61					
V8	1A7	60					
V9	2A0	17					
V10	2A1	67					
V11	2A2	20					
V12	2A3	70					
V13	2A4	19					
V14	2A5	69					
V15	2A6	68					
V16	2A7	18					
V17	1B0	65					
V18	1B1	14					
V19	1B2	64					
V20	1B3	11	0	61	nS	1-4	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXII. Output crosstalk, duration, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Fig
V21	1B4	12	0	61	nS	1-4	13
V22	1B5	62					
V23	1B6	63					
V24	1B7	13					
V25	2B0	24					
V26	2B1	74					
V27	2B2	23					
V28	2B3	73					
V29	2B4	22					
V30	2B5	72					
V31	2B6	71					
V32	2B7	21	0	61	nS	1-4	13
V33	-MASTER	94	0	45	nS	1-4	13
V34	READ	91	0	45	nS	1-4	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output crosstalk, duration, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 50 mV	V1-V34
Power supply current limit	0.894 A	NA	V1-V34
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 V1-V34 V1-V34
Output crosstalk measurement	 NA NA	 ± 3.05 nS ± 2.25 nS	 V1-V32 V33,V34

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

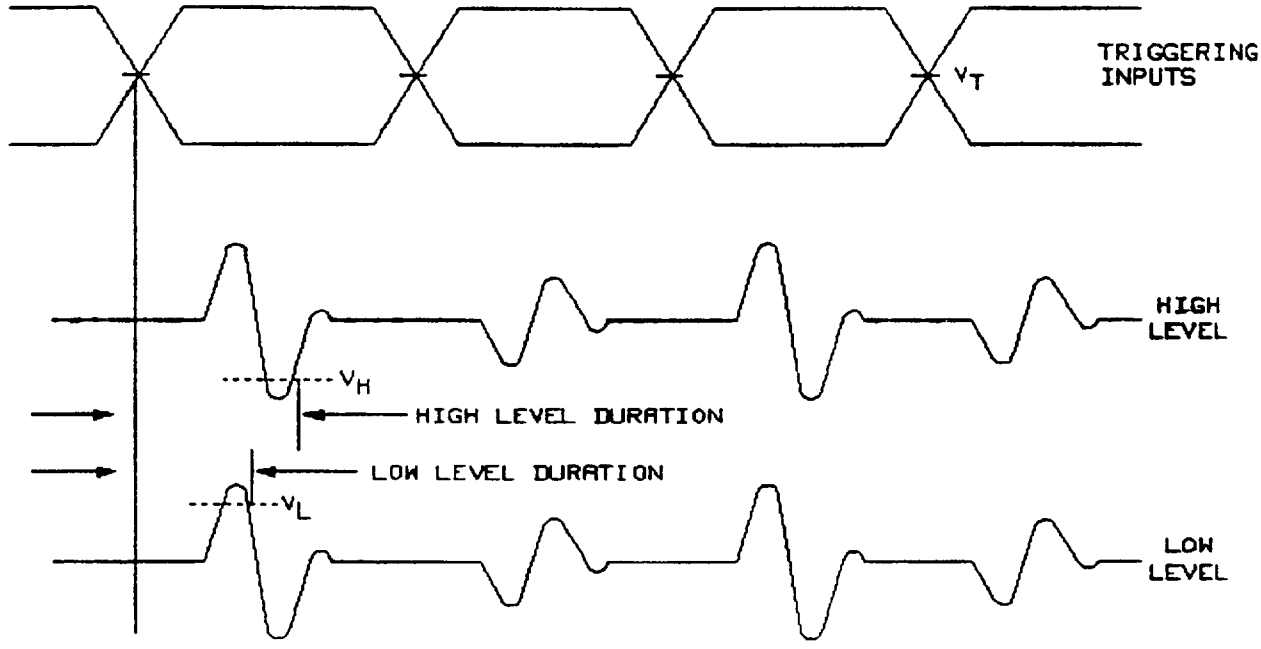
(d) For each output under test in table XXII, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 13.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 4.

4. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

1. VT = 1.3 V.
   VL = 0.4 V (AC coupled).
   VH = 0.4 V (AC coupled).

2. The duration is measured from the triggering edge of the input to the point where output crosstalk returns to the specified voltage.

FIGURE 13. Output crosstalk, duration waveforms for table XXII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	1A0	7	2	45	pF	1-3
W2	1A1	57				
W3	1A2	8				
W4	1A3	58				
W5	1A4	9				
W6	1A5	59				
W7	1A6	61				
W8	1A7	60				
W9	1B0	65				
W10	1B1	14				
W11	1B2	64				
W12	1B3	11				
W13	1B4	12				
W14	1B5	62				
W15	1B6	63				
W16	1B7	13				
W17	2A0	17				
W18	2A1	67				
W19	2A2	20				
W20	2A3	70	2	45	pF	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W21	2A4	19	2	45	pF	1-3
W22	2A5	69				
W23	2A6	68				
W24	2A7	18				
W25	2B0	24				
W26	2B1	74				
W27	2B2	23				
W28	2B3	73				
W29	2B4	22				
W30	2B5	72				
W31	2B6	71				
W32	2B7	21				
W33	-1OE	16				
W34	-2OE	75				
W35	1L/R	56				
W36	2L/R	66				
W37	BREQA	79				
W38	-BES	77	2	45	pF	1-3
W39	-DTACK	3	2	35	pF	1-3
W40	-1R	54	2	35	pF	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
W41	READ-B	36	2	35	pF	1-3
W42	-DACK1B	43				
W43	-DS-B	41	2	35	pF	1-3
W44	1C	55	2	25	pF	1-3
W45	CLK	53				
W46	WENI	51				
W47	-CSA	80				
W48	A0	37				
W49	BYTE	88				
W50	-DS-A	38				
W51	1D	4				
W52	-DACK2B	44				
W53	BREQB	95				
W54	-BACKB	45				
W55	-CSB	96	2	25	pF	1-3
W56	-CBREQIA	26	2	55	pF	1-3
W57	-CBREQIB	50				
W58	-DACK1A	78				
W59	-DACK2A	28				
W60	-BACKA	29	2	55	pF	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W61	READ-A	27	2	85	pF	1-3
W62	READ-S	85	2	85	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W62
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W62
Capacitance measurement	NA	± 2.15 pF	W1-W38
	NA	± 1.65 pF	W39-W43
	NA	± 1.15 pF	W44-W55
	NA	± 2.65 pF	W56-W60
	NA	± 4.15 pF	W61,W62

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Disable time test.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Figs
X1	-1OE to 1B0 (tPLZ)	56	Open	65	2.0	39.0	nS	14a,15a
X2	-1OE to 1B1 (tPLZ)			14				
X3	-1OE to 1B2 (tPLZ)			64				
X4	-1OE to 1B3 (tPLZ)			11				
X5	-1OE to 1B4 (tPLZ)			12				
X6	-1OE to 1B5 (tPLZ)			62				
X7	-1OE to 1B6 (tPLZ)			63				
X8	-1OE to 1B7 (tPLZ)			13				
X9	-1OE to 1B0 (tPHZ)			65				
X10	-1OE to 1B1 (tPHZ)			14				
X11	-1OE to 1B2 (tPHZ)			64				
X12	-1OE to 1B3 (tPHZ)			11				
X13	-1OE to 1B4 (tPHZ)	56	Open	12	2.0	39.0	nS	14a,15a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Disable time test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Figs
X14	-1OE to 1B5 (tPHZ)	56	Open	62	2.0	39.0	nS	14a,15a
X15	-1OE to 1B6 (tPHZ)	56	Open	63	2.0	39.0	nS	14a,15a
X16	-1OE to 1B7 (tPHZ)	56	Open	13	2.0	39.0	nS	14a,15a
X17	-1OE to 1A0 (tPLZ)	Open	56	7	2.0	39.0	nS	14b,15b
X18	-1OE to 1A1 (tPLZ)			57				
X19	-1OE to 1A2 (tPLZ)			8				
X20	-1OE to 1A3 (tPLZ)			58				
X21	-1OE to 1A4 (tPLZ)			9				
X22	-1OE to 1A5 (tPLZ)			59				
X23	-1OE to 1A6 (tPLZ)			61				
X24	-1OE to 1A7 (tPLZ)			60				
X25	-1OE to 1A0 (tPHZ)			7				
X26	-1OE to 1A1 (tPHZ)	Open	56	57	2.0	39.0	nS	14b,15b

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Disable time test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Figs
X27	-1OE to 1A2 (tPHZ)	Open	56	8	2.0	39.0	nS	14b,15b
X28	-1OE to 1A3 (tPHZ)			58				
X29	-1OE to 1A4 (tPHZ)			9				
X30	-1OE to 1A5 (tPHZ)			59				
X31	-1OE to 1A6 (tPHZ)			61				
X32	-1OE to 1A7 (tPHZ)	Open	56	60	2.0	39.0	nS	14b,15b
X33	-2OE to 2B0 (tPLZ)	66	Open	24	2.0	39.0	nS	14a,15a
X34	-2OE to 2B1 (tPLZ)			74				
X35	-2OE to 2B2 (tPLZ)			23				
X36	-2OE to 2B3 (tPLZ)			73				
X37	-2OE to 2B4 (tPLZ)			22				
X38	-2OE to 2B5 (tPLZ)			72				
X39	-2OE to 2B6 (tPLZ)	66	Open	71	2.0	39.0	nS	14a,15a

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Disable time test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Figs
X40	-2OE to 2B7 (tPLZ)	66	Open	21	2.0	39.0	nS	14a,15a
X41	-2OE to 2B0 (tPHZ)			24				
X42	-2OE to 2B1 (tPHZ)			74				
X43	-2OE to 2B2 (tPHZ)			23				
X44	-2OE to 2B3 (tPHZ)			73				
X45	-2OE to 2B4 (tPHZ)			22				
X46	-2OE to 2B5 (tPHZ)			72				
X47	-2OE to 2B6 (tPHZ)			71				
X48	-2OE to 2B7 (tPHZ)	66	Open	21	2.0	39.0	nS	14a,15a
X49	-2OE to 2A0 (tPLZ)	Open	66	17	2.0	39.0	nS	14b,15b
X50	-2OE to 2A1 (tPLZ)			67				
X51	-2OE to 2A2 (tPLZ)	Open	66	20	2.0	39.0	nS	14b,15b

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Disable time test - Continued.

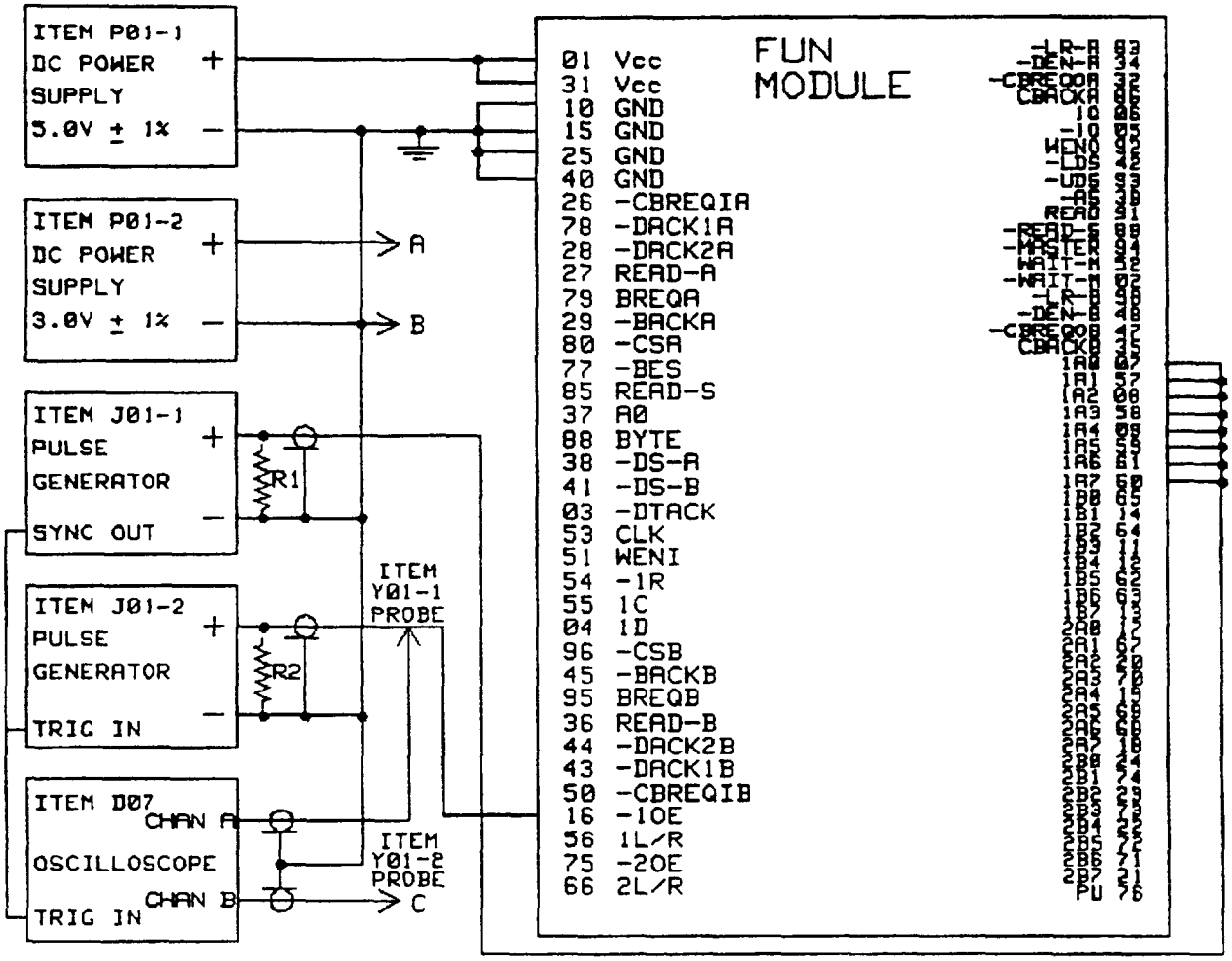
Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Figs
X52	-2OE to 2A3 (tPLZ)	Open	66	70	2.0	39.0	nS	14b,15b
X53	-2OE to 2A4 (tPLZ)			19				
X54	-2OE to 2A5 (tPLZ)			69				
X55	-2OE to 2A6 (tPLZ)			68				
X56	-2OE to 2A7 (tPLZ)			18				
X57	-2OE to 2A0 (tPHZ)			17				
X58	-2OE to 2A1 (tPHZ)			67				
X59	-2OE to 2A2 (tPHZ)			20				
X60	-2OE to 2A3 (tPHZ)			70				
X61	-2OE to 2A4 (tPHZ)			19				
X62	-2OE to 2A5 (tPHZ)			69				
X63	-2OE to 2A6 (tPHZ)			68				
X64	-2OE to 2A7 (tPHZ)	Open	66	18	2.0	39.0	nS	14b,15b

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Disable time test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Unit	Figs
X65	CBACKA to -AS (tPLZ)	95	29,26, 50,36, 37,88	39	0.0	33.0	nS	14c,15c
X66	CBACKA to -UDS (tPLZ)			93		32.0		
X67	CBACKA to -LDS (tPLZ)			42		32.0		
X68	CBACKB to READ (tPLZ)			91		32.0		
X69	CBACKA to -AS (tPHZ)			39		43.0		
X70	CBACKA to -UDS (tPHZ)			93		46.0		
X71	CBACKA to -LDS (tPHZ)			42		46.0		
X72	CBACKA to READ (tPHZ)	95	29,26, 50,36, 37,88	91	0.0	46.0	nS	14c,15c

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

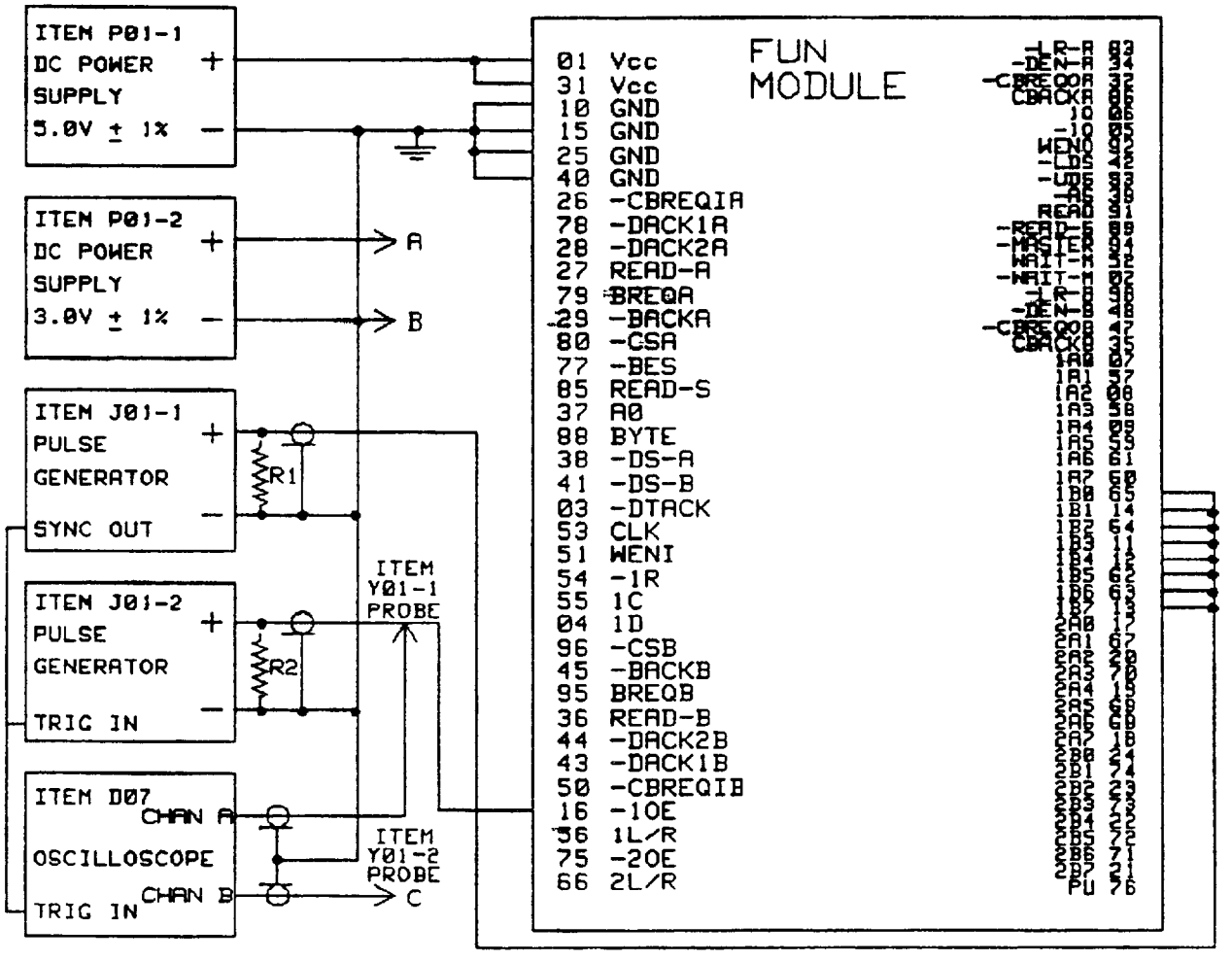
1. Output pins 65, 14, 64, 11, 12, 62, 63, 13, 24, 74, 23, 73, 22, 72, 71, and 21 shall be loaded as specified on figure 4, and the load power supply voltage shall be 2.1 V ± 1 percent.

2. R1 = R2 = 51 Ohms ± 5 percent, 1/2 W.

3. For test conditions X33-X48, Item J01-1 will be input for pins 17, 67, 20, 70, 19, 69, 68, and 18; Item J01-2 will be input for -2OE, pin 75.

FIGURE 14a. Disable time test circuit for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

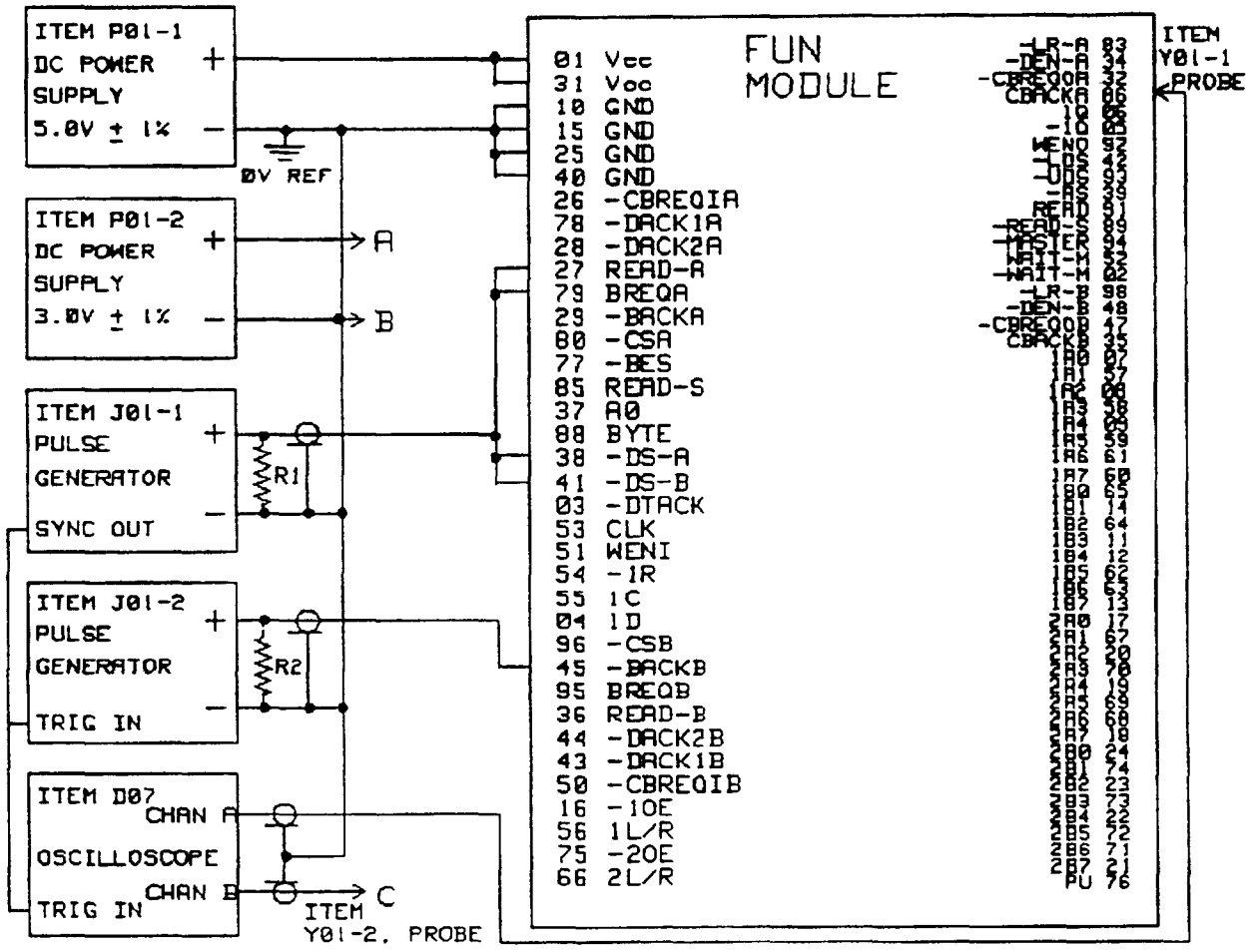
1. Output pins 7, 57, 8, 58, 9, 59, 61, 60, 17, 67, 20, 70, 19, 69, 68, and 18 shall be loaded as specified on figure 4, and the load power supply shall be 2.1 V ± 1 percent.

2. R1 = R2 = 51 Ohms ± 5 percent, 1/2 W.

3. For test conditions X49-X64, Item J01-1 will be input for pins 24, 74, 23, 73, 22, 72, 71, and 21; Item J01-2 will be input for -2OE, pin 75.

FIGURE 14b. Disable time test circuit for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

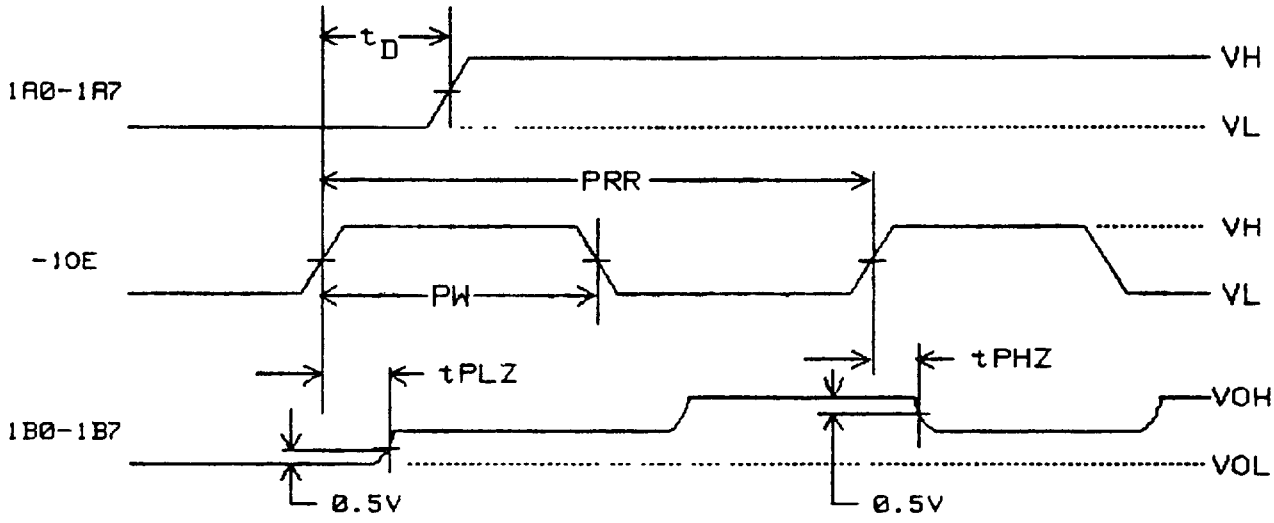
1. Output pins 39, 42, 91 and 93 shall be loaded as specified on figure 4, and the load power supply shall be 2.5 V ± 1 percent.

2. R1 = R2 = 51 Ohms ± 5 percent, 1/2 W.

3. Item J01-1 shall be in double pulse mode.

FIGURE 14c. Disable time test circuit for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

1. PW = 250 nS ± 25 nS.

2. PRR = 500 nS ± 50 nS.

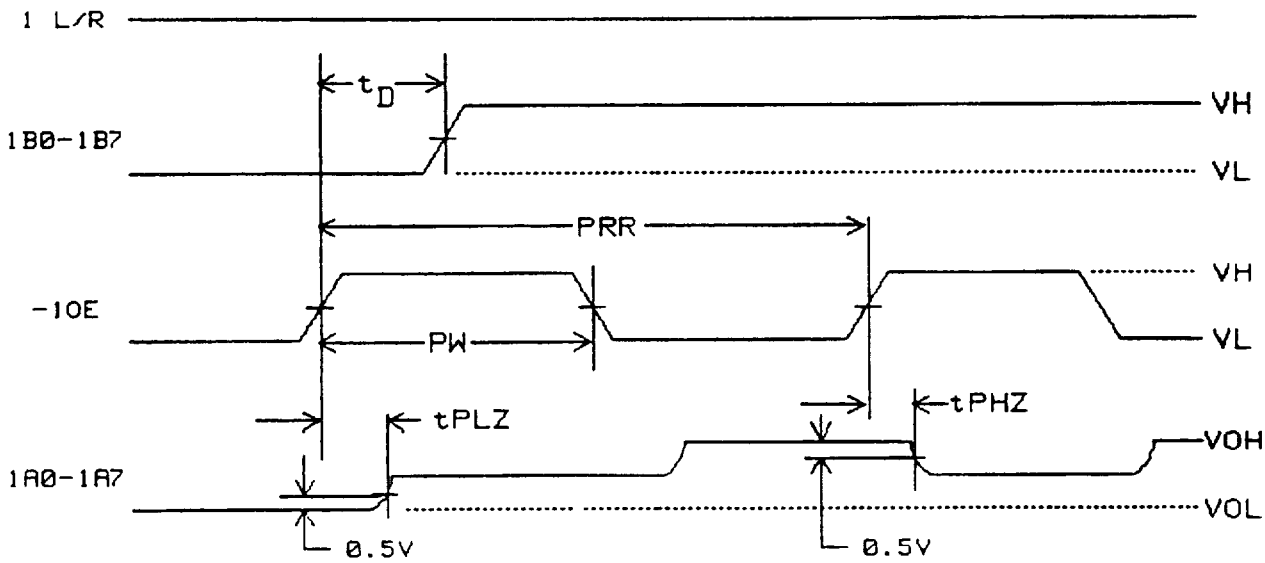
3. tD = 180 nS ± 18 nS.

4. Transition times shall be ≤ 6 nS.

5. For test conditions X33-X48, inputs are 2A0-2A7 and outputs are 2B0-2B7.

FIGURE 15a. Disable time test circuit waveform for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

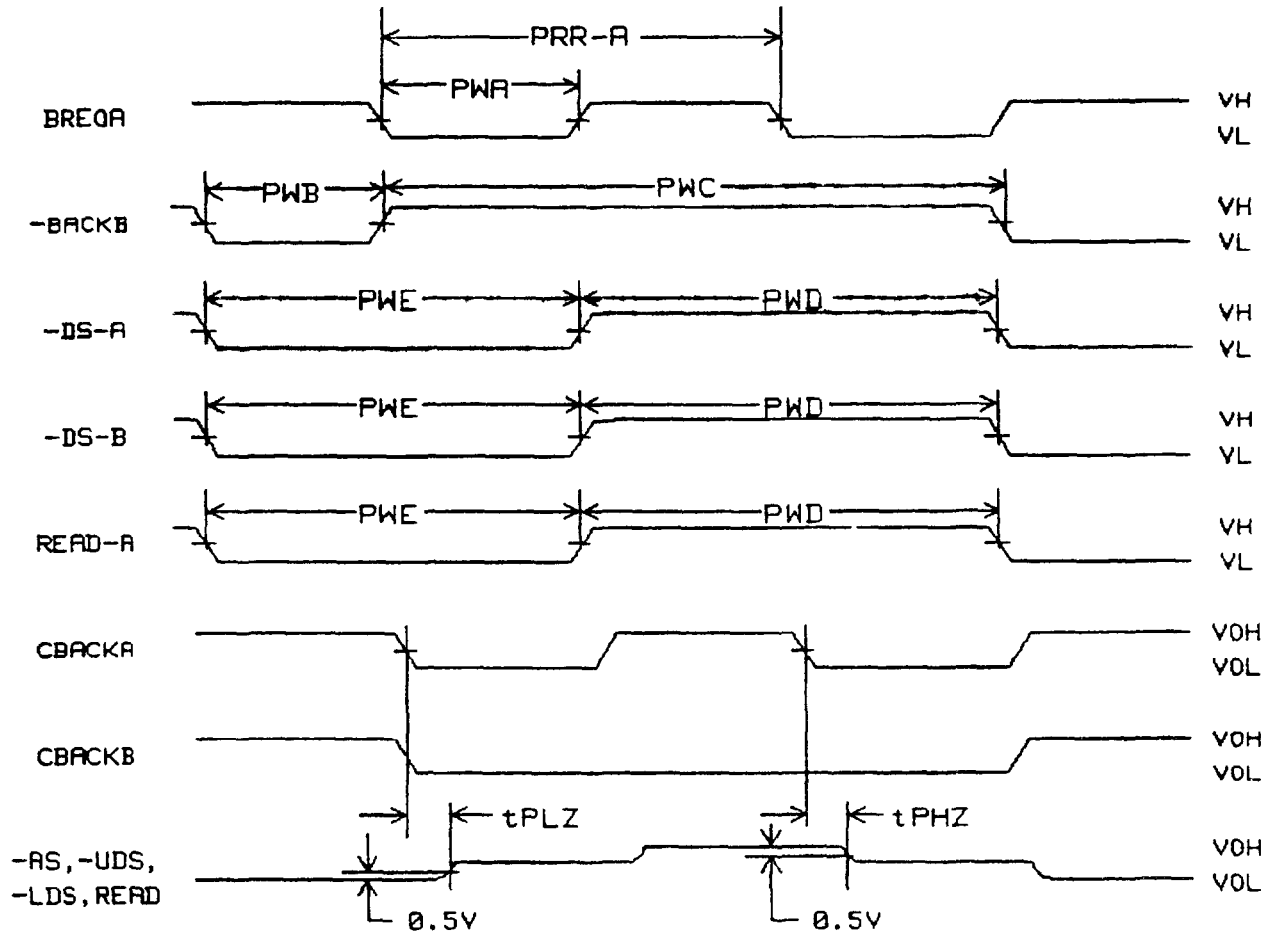


NOTES:

1. PW = 250 nS ± 25 nS.
2. PRR = 500 nS ± 50 nS.
3. tD = 180 nS ± 18 nS.
4. Transition times shall be ≤ 6 nS.
5. For test conditions X49-X64, inputs are 2B0-2B7 and outputs are 2A0-2A7.

FIGURE 15b. Disable time test circuit waveform for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.



NOTES:

1. PWA = 250 ± 25 nS.
   PRR-A = 500 ± 50 nS, Item J01-1.

2. PWB = 250 ± 25 nS.
   PWC = 750 ± 75 nS, Item J01-2.

3. PWD = PWE = 500 ± 50 nS, Item J01-1.

4. Transition times shall be ≤ 6 nS.

FIGURE 15c. Disable time test circuit waveform for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

TABLE XXV. Threshold test.

Test condition	Test description	Notes
Y	Threshold verification	1-3

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply	4.5 V	± 45 mV	Y
Power supply current limit	894 mA	NA	Y
Pattern voltage high, (VH)	2.0 V	± 20 mV	Y
low, (VL)	0.7 V	± 7 mV	Y
low, (VL)	0.8 V	± 8 mV	Y
Output voltage high, (Voh)	NA	± 125 mV	Y
low, (Vol)	NA	± 190 mV	Y
low, (Vol)	NA	± 185 mV	Y

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results with the guaranteed maximum low level voltage (0.5 V ± 5.5 mV) and the guaranteed minimum high level voltage (2.4 V ± 23.1 mV). The module inputs shall not violate the minimum input timing requirements shown on figure 7.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Abbreviations used are defined in 6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX

MODULE FUNCTION

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

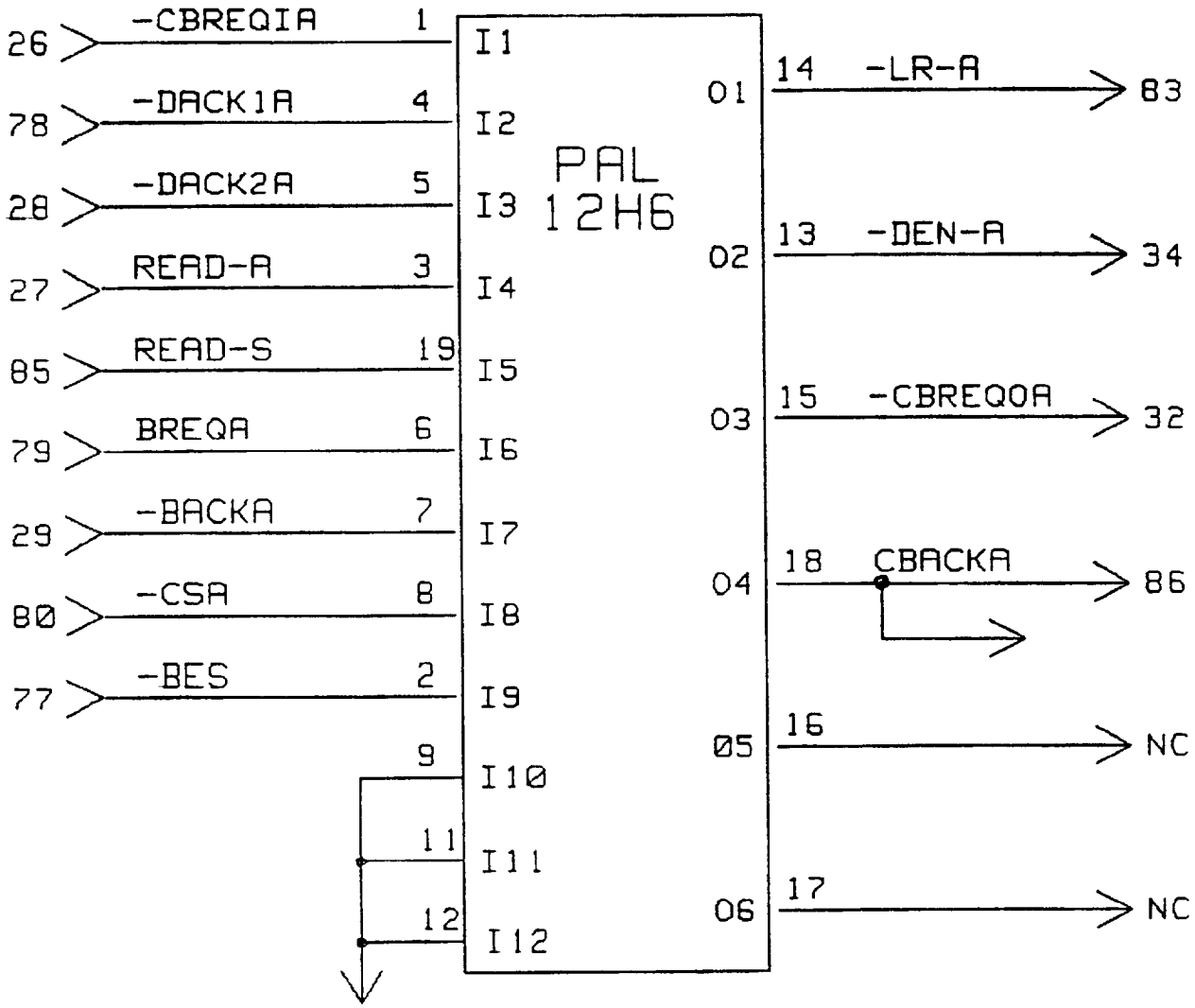
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX

30. Module function. The FUN module is located in the control timing and interface (CTI) and I/O partitions of the Standard Computing Element (SCE). The module is designed to provide support for two FUM DMA controller modules which are built around the AM9516 DMA controller. The FUN module serves as an interface between the FUM DMA controller and a subsystem driven by a 68000 microprocessor module (NAH). The FUN module also conditions signals between the DMA controller and the MPU communications support module (AHH), as well as conditioning subsystem control signals. The module contains two DMA controller operation slave logic blocks (see figure 1). Each is designed to support a FUM DMA controller module during slave operation. The -LR-A and -LR-B outputs are provided to the DMA controller to control the direction of data flow. The module contains a DMA controller master logic block (see figure 1) that, during master operation, generates subsystem control signals from DMA controller inputs. The operation slave and master logic blocks are implemented with PAL's (refer to figures 1 to 6). The FUN module supplies a wait signal (WAIT-M) to the DMA controller to allow extension of the DMA transfer operation for slow memory or peripherals. Two identical general purpose 8-bit transceiver logic blocks (2-54LS245) are available in the FUN module. The input enable signals (-1OE,-2OE) allow the outputs to be enabled or placed in the high impedance state. The direction inputs (1L/R, 2L/R) control the direction of signal flow through the transceivers. When XL/R is logic 1, signal flow is from XA0-XA7 to XB0-XB7. When XL/R is a logic 0, signal flow is from XB0-XB7 to XA0-XA7 (refer to figure 7 and table I). A D Flip-Flop (54LS74) is also fully pinned out on the module and is available for general purpose use. One independent pull-up resistor (1 KOhm) is provided for general purpose use. Pull-up resistors (6.04 KOhm) are provided on the 32 transceiver I/O pins so that all transceiver I/O pins are pulled to a valid logic high when the transceivers are in a high impedance state. The pull-up resistors prevent oscillation on the noise sensitive high impedance transceiver signal lines.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX



$-CBREQOA = \overline{BREQA} + (\overline{-BACKA} * -CBREQIA)$

$CBACKA = \overline{-CBREQIA} * BREQA * \overline{-BACKA}$

$-LR-A = (\overline{-BACKA} * \overline{READ-A} * -DACK1A * -DACK2A) + (-BACKA * \overline{-CSA} * READ-S)$

$-DEN-A = \overline{-BACKA} + -CSA + -BES$

FIGURE 1. FUN module control PAL - channel A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX

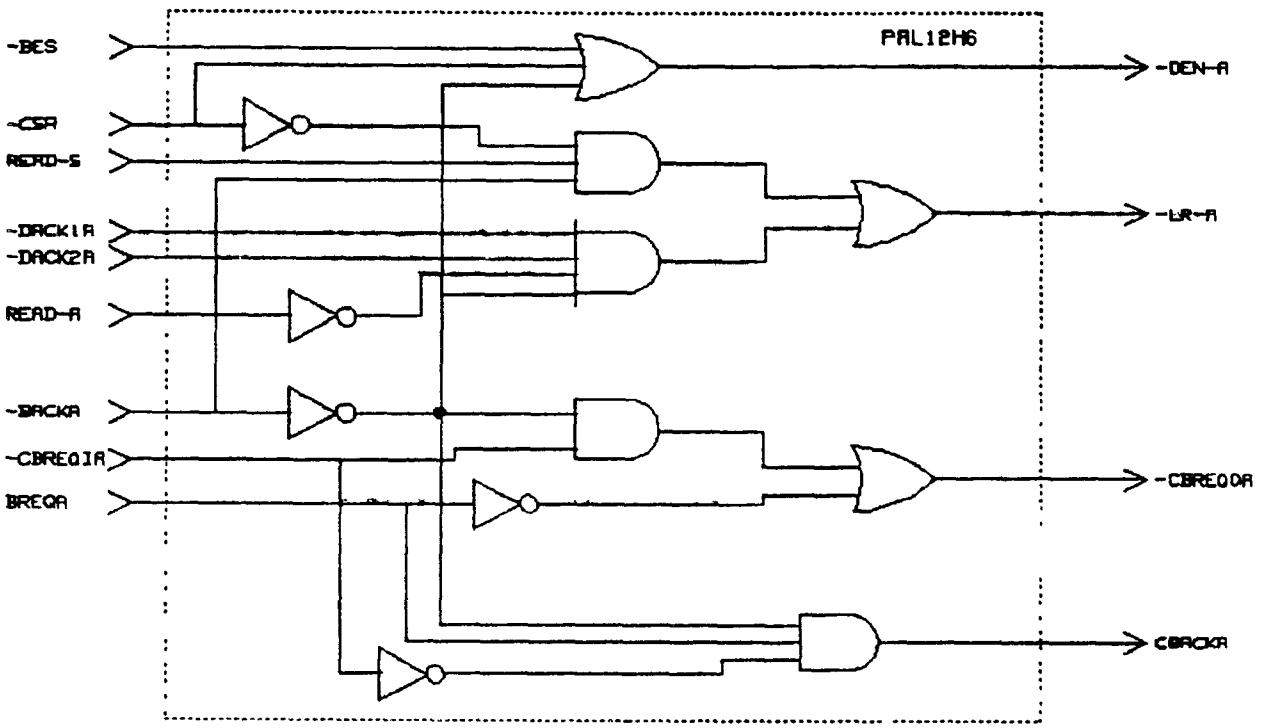
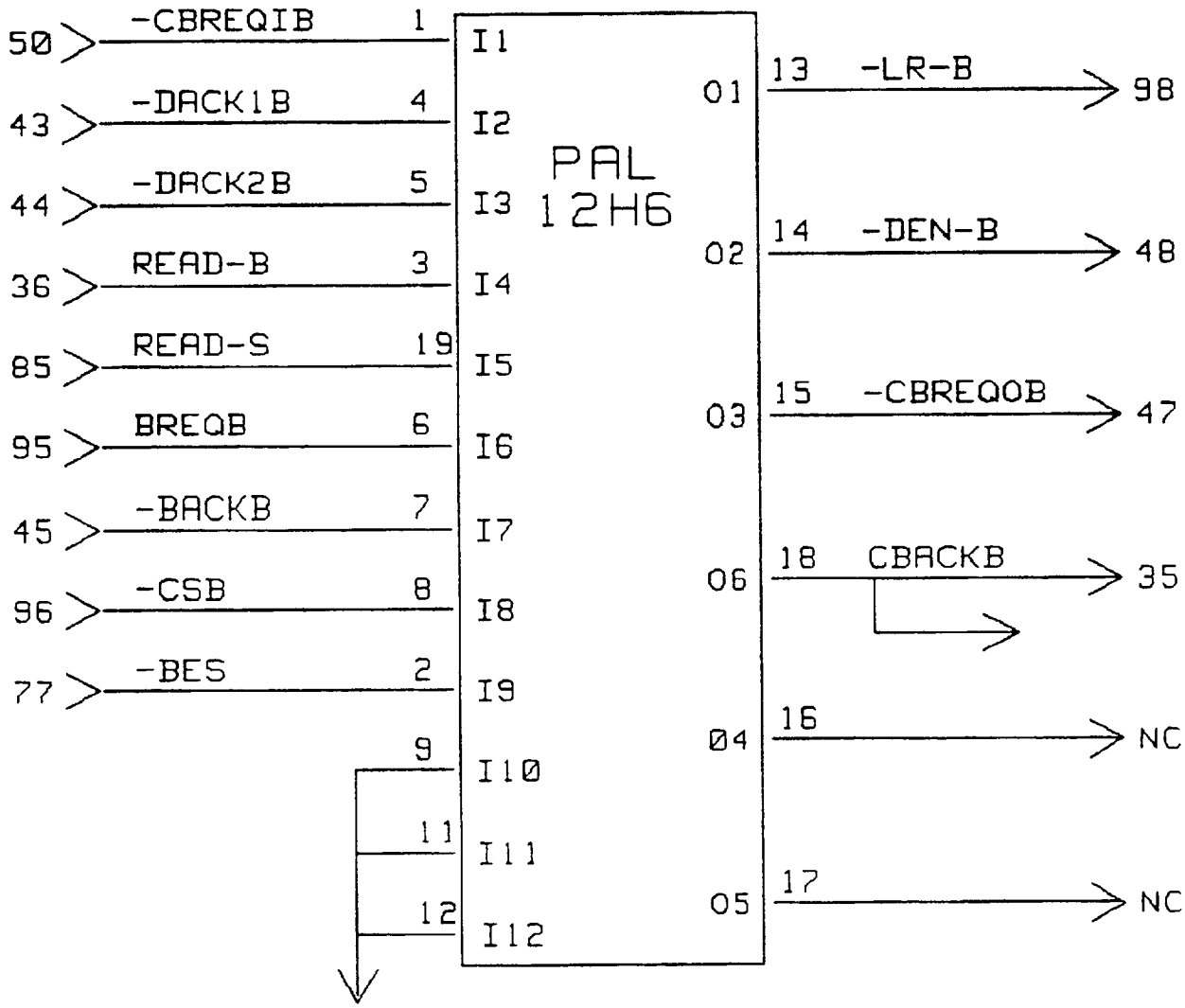


FIGURE 2. PAL internal logic diagram - channel A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX



$-CBREQOB = \overline{BREQB} + (\overline{-BACKB} * -CBREQIB)$

$CBACKB = \overline{-CBREQIB} * BREQB * \overline{-BACKB}$

$-LR-B = (\overline{-BACKB} * \overline{READ-B} * -DACK1B * -DACK2B) + (-BACKB * \overline{-CSB} * READ-S)$

$-DEN-B = \overline{-BACKB} + -CSB + -BES$

FIGURE 3. FUN module control PAL - channel B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX

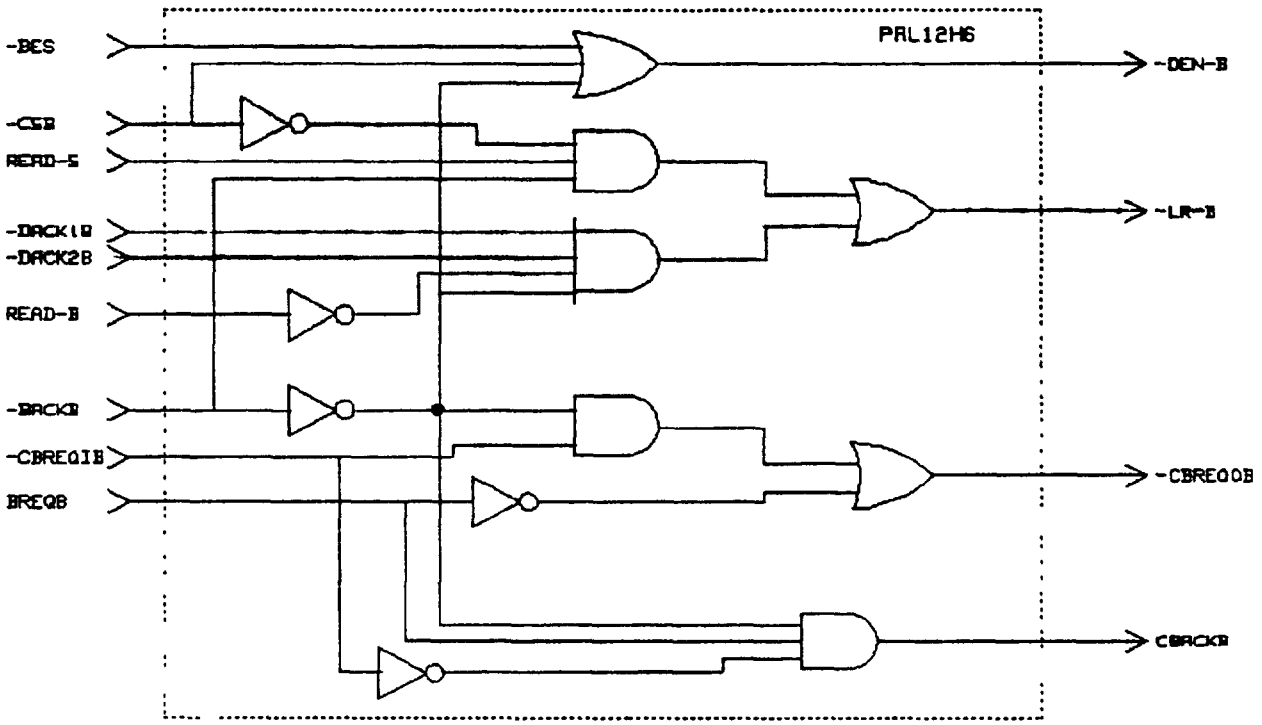
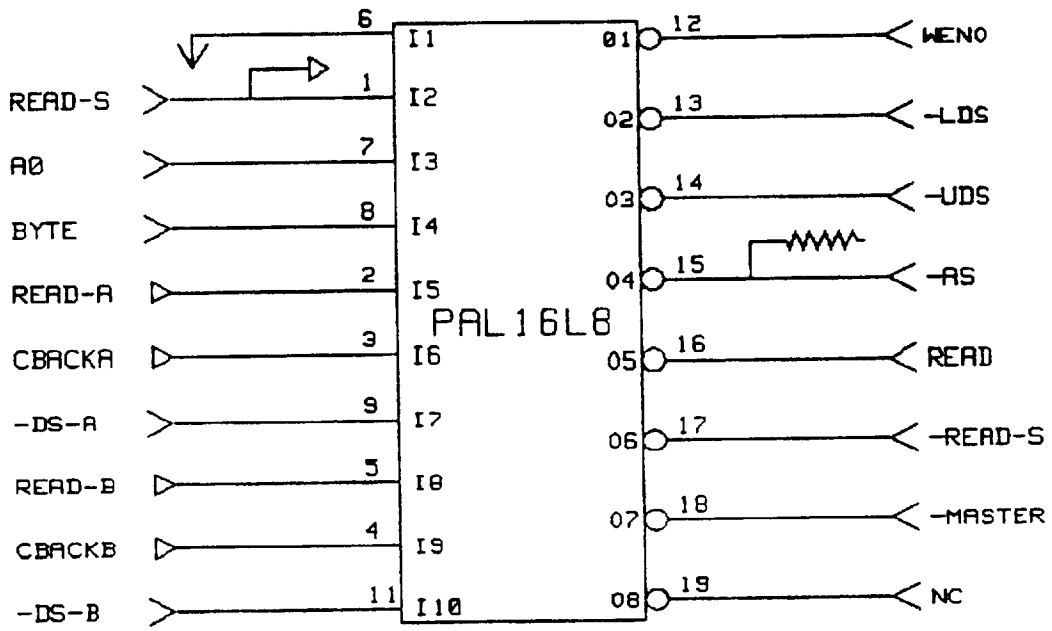


FIGURE 4. PAL internal logic diagram - channel B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX



If (MASTER) READ = CBACKA * $\overline{\text{READ-A}}$ + CBACKB * $\overline{\text{READ-B}}$

If (MASTER) -LDS = (CBACKA * $\overline{\text{-DS-A}}$ $\overline{\text{A0}}$ + CBACKB * $\overline{\text{-DS-B}}$ * $\overline{\text{A0}}$
+ CBACKA * $\overline{\text{BYTE}}$ * -DS-A + CBACKB * $\overline{\text{BYTE}}$ * -DS-B)

If (MASTER) -UDS = (CBACKA * $\overline{\text{-DS-A}}$ $\overline{\text{A0}}$ + CBACKB * $\overline{\text{-DS-B}}$ * $\overline{\text{A0}}$
+ CBACKA * $\overline{\text{BYTE}}$ * -DS-A + CBACKB * $\overline{\text{BYTE}}$ * -DS-B)

If (MASTER) -AS = CBACKA * $\overline{\text{-DS-A}}$ + CBACKB * $\overline{\text{-DS-B}}$

If (MASTER): READ,-LDS,-UDS,-AS = High impedance.

WENO = -DS-A * -DS-B

-READ-S = $\overline{\text{READ-S}}$

-MASTER = $\overline{\text{CBACKA + CBACKB}}$

FIGURE 5. FUN module control PAL - master channel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX

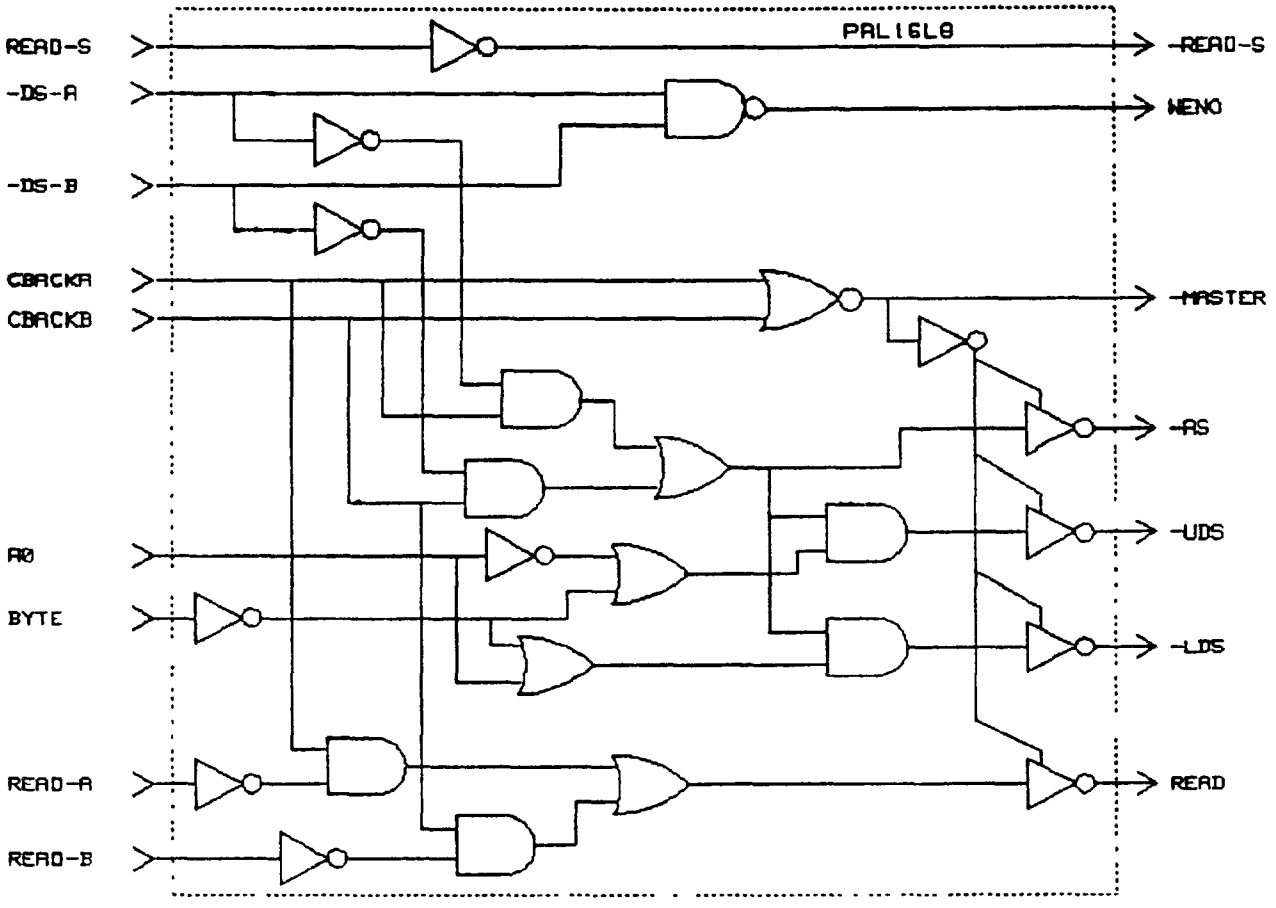


FIGURE 6. PAL internal logic diagram - master channel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

APPENDIX

TABLE I. Input to output relationship for buffers.

Input		Operation
Enable (-XOE)	Direction control (XL/R)	
L	L	B data to A bus
L	H	A data to B bus
H	X	Isolation

1/ H = high level
2/ L = low level
3/ X = irrelevant

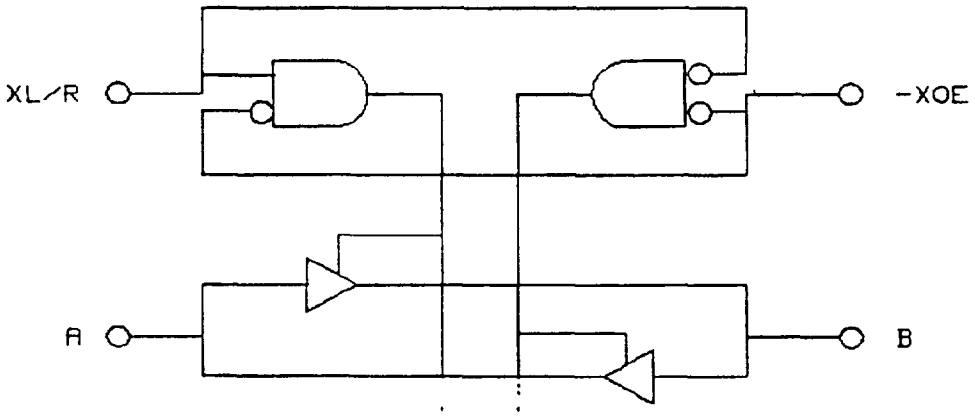


FIGURE 7. Logic diagram of bidirectional buffer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-W-28787/358	2 DOCUMENT DATE (YYMMDD) 1990 October 26

3 DOCUMENT TITLE
Modules, Digital, Standard Electronic DMA Support Module, KeyCode Fun

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b TELEPHONE (Include Area Code) (1) Commercial TPOC: 812-854-1528	(2) AUTOVON 88-482-1528
c ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN. Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.